

Further Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



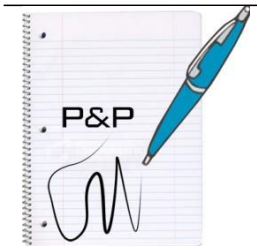
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



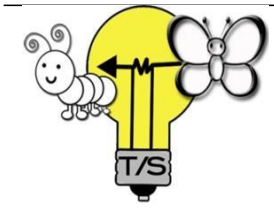
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: you would do "3 times 2 is 5 ... let me check – no is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: ——— then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: —

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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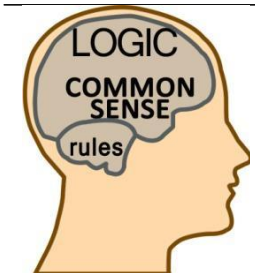
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

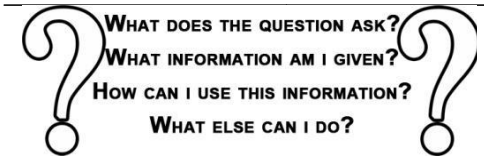
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Table of Contents

CHAPTER 1: Data Analysis **5**

Exercise 1: Data Types & Representation 6

Exercise 2: Summary Statistics 11

Exercise 3: Normal Distribution 16

Exercise 4 Box Plots 19

Exercise 5: Correlation 25

CHAPTER 2: Number Patterns 29

Exercise 1: Arithmetic Sequences 30

Exercise 2: Geometric Sequences 32

Exercise 3: Sum to Infinity 35

Exercise 4: Difference Equations 38

CHAPTER 3: Geometry & Trigonometry **41**

Exercise 1: Pythagoras' Theorem 42

Exercise 2: Similarity 48

Exercise 3: Volume & Surface Area 54

Exercise 4: Change of Scale 60

Exercise 5: Trigonometry (I) 63

Exercise 6: Trigonometry (II) 69

CHAPTER 4: Graphs & Relations **74**

Exercise 1: Linear Relationships 75

Exercise 2: Simultaneous Equations 80

Exercise 3: Non-linear Relationships 83

Exercise 4: Proportional Relationships 88

Exercise 5: Linear Programming 90

CHAPTER 5: Networks **94**

Exercise 1: Representation of Networks 95

Exercise 2: Trees 100

Exercise 3: Paths & Flow 105

Exercise 4:Optimisation 109

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CHAPTER 6: Matrices 114

Exercise 1: Representation & Operations 115

Exercise 2: Simultaneous Equations 120

Exercise 3: Transition Matrices 123

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Further Mathematics

Data Analysis

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Exercise 1

Data Types & Representation

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1) For which of the following would all data be available for analysis, and which would require a sample to be taken?

a) Score distribution in a basketball competition

b) Voting intentions of the Australian people

c) Favourite colour of your class

d) Favourite car of the people of Sydney

e) Types of dogs owned by the people of Victoria

2) Classify the following data as either quantitative or categorical. If the data is quantitative, indicate if it is discrete or continuous

a) Heights of your class members

b) Attendance at football games

c) Car colours

d) Dog breeds

e) Courses offered at a university

f) Number of people enrolled in each course at a university

3) Construct a frequency histogram of the following data

12, 10, 15, 8, 7, 12, 8, 16, 21, 22, 21, 12, 10, 8, 22, 21, 15, 11, 12, 22, 12, 16, 21, 8, 10, 8, 15, 9, 23, 17, 67, 7, 8, 16, 12, 21, 14, 15, 10

4) Construct a cumulative frequency histogram from the following data of the weights of 30 people in a group (in kg)

72, 73, 73, 75, 77, 77, 78, 80, 83, 84, 84, 84, 85, 85, 88, 88, 90, 92, 92, 93, 95, 95, 96, 97, 97, 98, 98, 100, 103, 104

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5) The following data shows the time taken for the members of an athletic club to run 100 metres

12.2, 12.4, 13.1, 13.2, 13.3, 13.4, 13.4, 13.5, 13.8, 14.1, 14.2, 14.3, 15, 15.2, 15.5, 15.5, 15.7, 15.8, 16, 16.2

a) Group the data into class intervals

b) Construct a histogram of the grouped data

c) Find the mean of the grouped data

d) Find the modal class of the grouped data

6) Organise each of the following data sets into stem and leaf plots

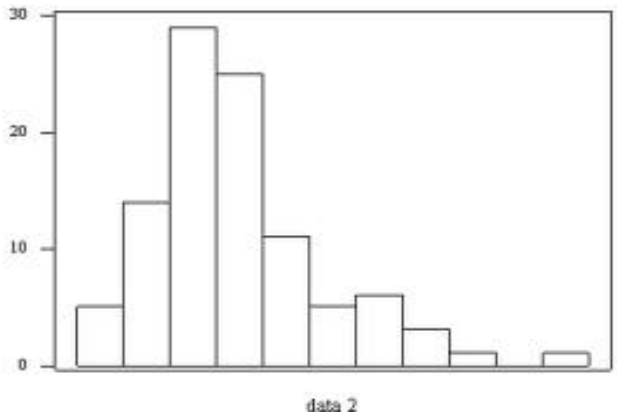
a) 20, 23, 25, 31, 32, 34, 42, 42, 43, 26, 37, 41, 30, 25, 26, 53, 27, 33, 23, 30, 41

b) 73, 62, 66, 76, 78, 80, 83, 99, 92, 75, 74, 88, 99, 70, 71, 69, 66, 73, 81

c) 12, 10, 22, 24, 35, 46, 47, 32, 31, 43, 22, 21, 45, 56, 43, 32, 37, 49, 40, 21, 20, 30, 27, 26, 32, 21, 50, 60, 22

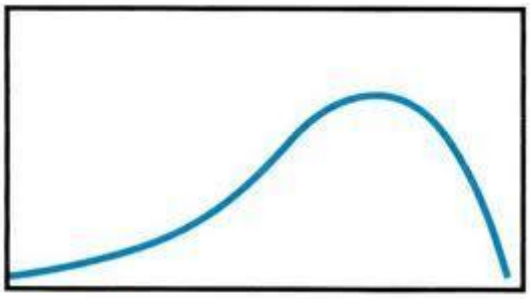
7) Describe the following graphs in terms of skewness

a)

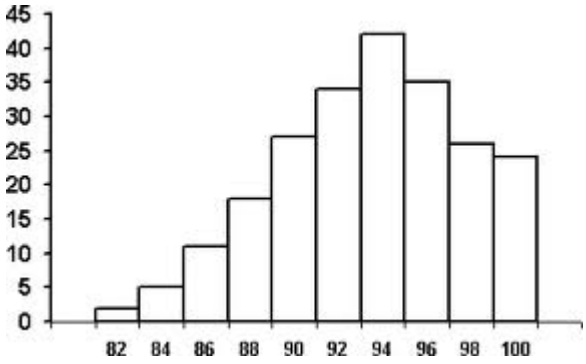


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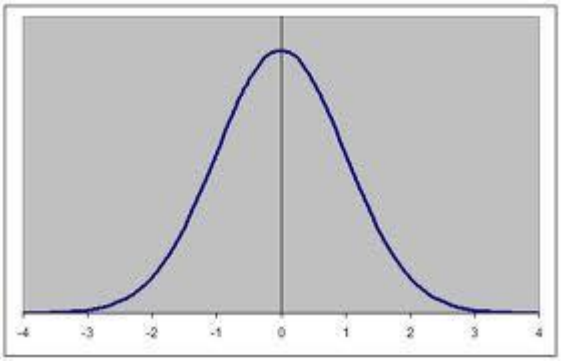
b)



c)



d)



8) On Monday Tom spent $5 on lunch, $10 on petrol, $20 on clothes, and $25 on music. On Tuesday he spent $15 on lunch, $15 on petrol, $10 on clothes, and $10 on music. On Wednesday he spent $10 on lunch, $40 on petrol, $5 on clothes, and $30 on music. Represent the above data in a percentage segmented bar chart

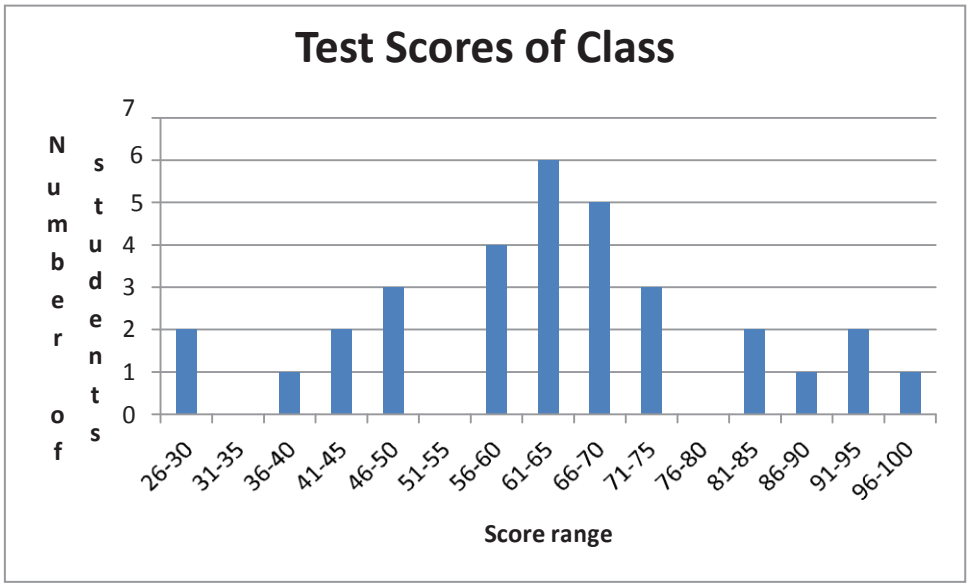
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9) The following data shows the pulse rate of a number of patients

68 60 76 68 64 80 72 76 92 68 56 72 68 60 84, 72 56 88 76 80 68 80 84 64 80 72 64 68 76 72

Represent the data in a dot plot, and determine the range of the data. Discuss one disadvantage of using dot plots to analyse data in terms of range.

10) Construct a cumulative frequency table from the following bar graph



How many students sat the test, and how many passed?

11) The following data set is the set of scores of football team A during its season

34, 38, 42, 43, 45, 48, 49, 51, 53, 57, 58, 60, 61, 63, 67, 71, 74, 77, 79, 85

The following data set is the set of scores of football team B during its season

23, 29, 35, 39, 46, 47, 49, 52, 53, 53, 59, 67, 73, 79, 86, 91, 97, 101, 117, 126

Display the data in a back to back stem and leaf plot

What were the respective median scores, and which team was more consistent during the season

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Exercise 2

Summary Statistics

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1) Find the mean, mode & median of the following data sets

a) 10, 7, 5, 7, 6, 3, 4, 3, 20, 7, 6, 6, 15, 14, 7

b) 4, 20, 8, 13, 12, 15, 8, 15, 18, 7, 13, 9, 20, 17, 1

c) 4, 19, 20, 16, 11, 16, 1, 10, 15, 5, 18, 17, 19, 14, 4

d) 12, 8, 2, 4, 7, 2, 1, 9, 16, 15, 17, 1, 1, 20, 14

e) 17, 5, 3, 15, 19, 12, 5, 1, 3, 11, 18, 17, 14, 1, 7

2) Find the mean mode and median from the following frequency distribution tables

a)

Value	Frequency
1	6
2	4
3	1
4	0
5	5
6	4
7	7

b)

Value	Frequency
20	2
21	2
22	2
23	3
24	2
25	4
26	3

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c)

Value	Frequency
11	1
12	1
13	1
14	1
15	1
16	1
17	1

d)

Value	Frequency
1	6
2	4
3	1
4	0
5	5
6	4
7	7
1000	1

3) Using your answers to parts 2a and 2d, what effect does an outlier have on the value of the mode, mean & median?

4) Represent the following test scores in a stem and leaf plot, and use it to calculate the mean, mode, median & range of the data

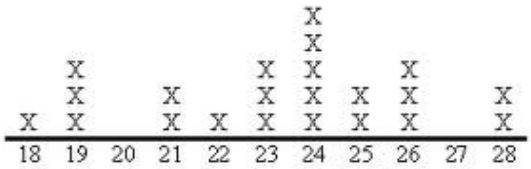
a) 83, 80, 48, 71, 61, 58, 47, 52, 56, 78, 86, 47, 62, 57, 77, 60, 46, 89, 81, 72

b) 48, 88, 50, 49, 54, 56, 57, 47, 48, 84, 62, 82, 69, 79, 51, 48, 89, 49, 65, 75

c) 74, 84, 69, 61, 79, 81, 77, 56, 50, 48, 51, 61, 90, 76, 53, 47, 56, 52, 89, 88

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5) Calculate the mean, mode, median & range for the following dot plot



6) The mean of a set of data is 25, its mode is 30 (there are 10 scores of 30), and its median is 28. A new score of 200 is added to the set. What effect will this new score have on the mean, mode & median?

7) Fifteen students sat a maths test and their mean mark was 60%. Alan was sick for the test and sat it later. When his score was added to the data set, the mean mark had increased to 62%. What score did Alan get on the test?

8) There are 15 girls and 15 boys in a class. On a test the girls mean mark was 80%, while the mean mark of the boys was 70%. What was the mean mark for the class?

9) There are 20 girls and 10 boys in a class. On a test the girls mean mark was 80% while the mean mark of the boys was 70%. What was the mean mark for the class?

10) Why are the answers to questions 6 and 7 different, given that the mean marks of the boys and girls in both classes were the same?

11) The following set of data is in order. Its mean is 30 and its median is 14. What are the values of x and y?

5, 8, x, 12, y, 40, 50, 100

12) Find the range of the following sets of data

a) 1, 2, 5, 7, 10

b) 3, 6, 18, 19, 100

c) 1, 1, 1, 1, 1

d) 17, 3, 18, 22, 30, 4, 10

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e) 40, 30, 20, 10, 0

f) -5, 7, 15, 22, 40, 51

13) Find the inter-quartile range of the following data sets

a) 7, 15, 20, 22, 25, 32, 40

b) 1, 5, 6, 12, 20, 30, 50

c) 2, 10, 18, 24, 32, 80, 82, 90

d) 23, 25, 4, 12, 21, 50, 32, 43, 5, 60, 45

15) Can the inter-quartile range be less than the range for a set of data? Explain

16) Can the inter-quartile range be equal to the range for a set of data? Explain

17) What is the standard deviation of the following sets of data?

a) 2, 2, 2, 2, 2, 2

b) 1, 2, 3, 4, 5

c) 3, 6, 9, 12, 15

d) 4, 20, 40, 60, 100

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Exercise 3

Normal Distribution

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1) Describe what the following z values tell us about the data point in relation to the mean

a) $z = 0$

b) $z = 1$

c) $z = -2$

d) $z > 2$

2) Calculate the z score of a score of 8 in a data set that has a mean of 6 and a standard deviation of 2. Describe the position of the data point in relation to the mean

3) A data point has a z score of 1.5. The data set has a mean of 5 and a standard deviation of 3. What is the data point?

4) A data set has a mean of 17.5. The data point 33.5 is 1.6 standard deviations from the mean. What is the value of the standard deviation?

5) The data point 41 lies within a set of data having a standard deviation of 6. If the data point is 4 standard deviations from the mean, what is the value of the mean?

6) If a set of data is normally distributed what percentage of the scores are within 1 standard deviation from the mean?

7) 95% of people in a group are between 77kg and 103 kg. What is the mean and standard deviation if we assume the data is normally distributed?

8) A teacher gives a maths test with the pass mark being 25 out of 50. The class scores the following marks:

12, 14, 10, 22, 35, 38, 13, 22, 40, 11, 22, 24, 25, 30, 5, and 18

The teacher sees that the majority of the class will fail the test, and he decides to standardise the marks. He will only fail a student that is more than one standard deviation below the mean

How many students now pass the test?

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9) Another teacher is determining the term marks for his class and wants to grade according to the following formula

Standard Deviations from mean	Grade
Score ≥2 s.d.	A
1 s.d. ≤ score < 2 s.d.	B
0 s.d. ≤ score < 1 s.d.	C
-1 s.d. ≤ score < 0 s.d.	D
Score< -1 s.d	E

Grade the following students

NAME	SCORE
James	62
Mark	38
Karen	84
Janine	70
Carol	65
June	68
Peter	44
Kevin	48
Brian	56
Alan	66
Bree	53

10) Deliveries of sand made by a nursery are advertised as 100 kg. The mean of the deliveries is 100 kg with a standard deviation of 1.2 kg

a) Within what weight range will 95% of the deliveries be?

b) What percentage of deliveries will be between 100 kg and 101.2 kg?

c) The company offers money back if any of the deliveries are 3 or more standard deviations below the mean. If they made 5000 deliveries in one month, how many of these will have to be refunded?

(Assume the data is normally distributed)

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Exercise 4

Box Plots

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1) The top ten test batting averages in history are:

99.94, 60.97, 60.83, 60.73, 59.23, 58.67, 58.61, 58.45, 57.78, 57.02

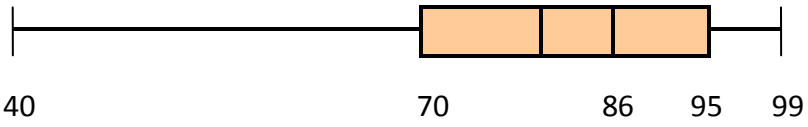
a) Construct a box and whisker plot of the data

b) Predict if the mean will be higher or lower than the median and justify your answer

c) Calculate the mean and standard deviation

Is the range, inter-quartile range or standard deviation a more realistic measure of the spread of the data in this case? Justify your answer

2) The box and whisker plot below shows the distribution of students' maths test scores



a) .What was the lowest score in the test?

b) What percentage of the class scored above 70%?

c) What was the median score?

d) What percentage of the class scored between 70 and 86?

e) Comment on the difficulty of the test

3) The following data shows the score (in points) of the winning AFL football teams over three weeks

61, 63, 72, 75, 80, 84, 84, 86, 90, 96, 97, 102, 105, 105, 106, 107, 108, 110, 111, 115, 120, 122, 125, 130

a) Draw a box-and-whisker plot marking the 5 relevant points

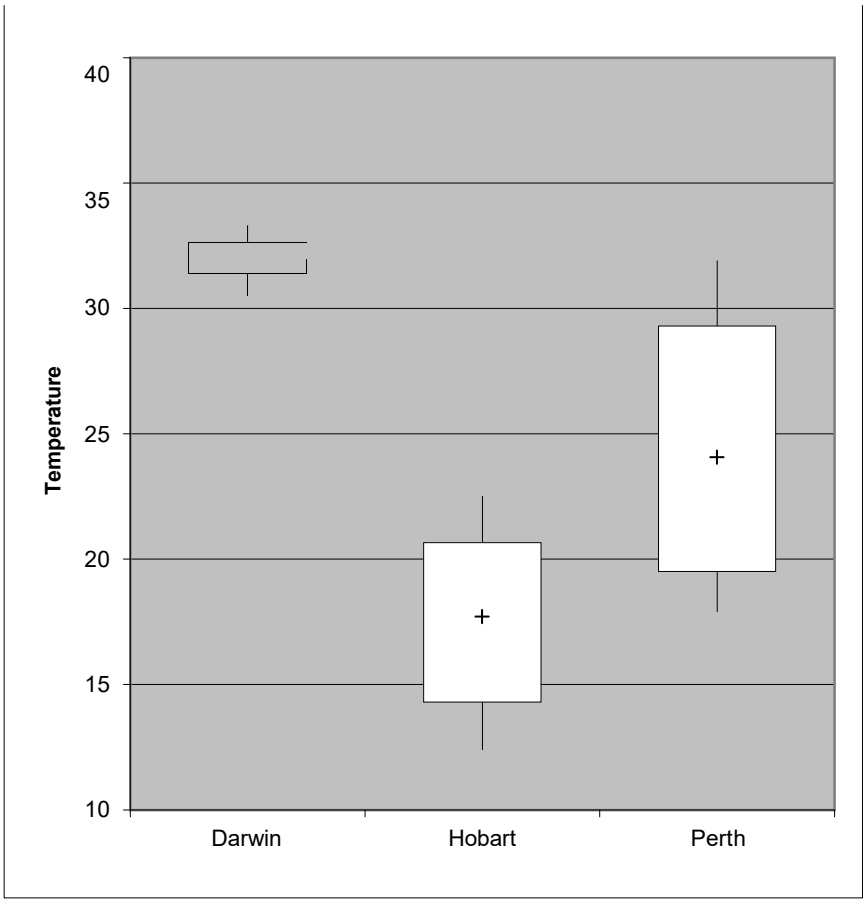
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b) What is the inter-quartile range?

c) What is the median score?

d) Comment on the spread of the data

4) The following box plots show the distribution of the average monthly temperatures for a year for Hobart, Darwin and Perth. (the cross indicates the median for each data set)



Compare the three sets of data and comment on the similarities and differences in the distributions of average monthly temperatures for the three cities

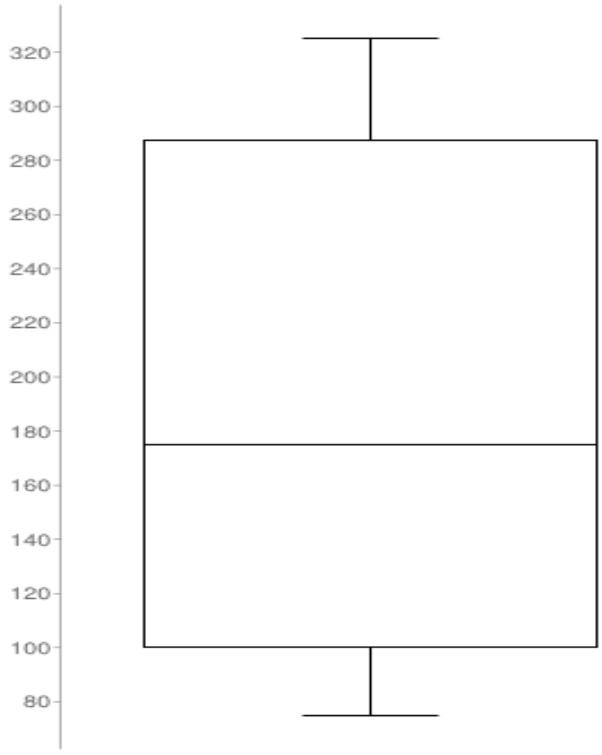
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5) The following data is in stem and leaf form. Represent it as a box and whisker plot

Stem	Leaf
2	2 4 7 7 9
3	0 1 1 1 3 3 5 6
4	5 5 5 7 8 9
5	1 1 2 2 5
6	3 3 5 6 7
9	5

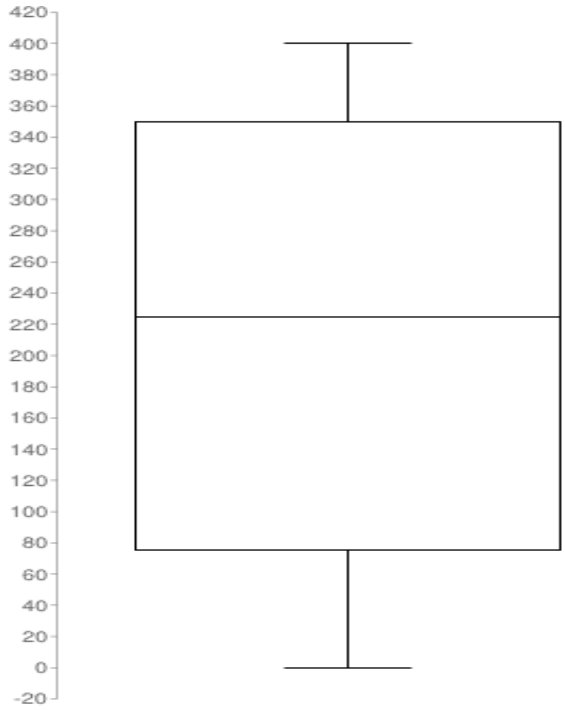
6) Draw a box and whisker plot for a set of data that has a median of 20, an inter quartile range of 15, and a range of 40

7) The following box plot shows the distribution of the average rainfall for Great Lake for the past 40 years



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The following box plot shows the same data set for Water World



a) Which site has the greater median average rainfall?

b) Which site has the record lowest annual rainfall and record highest annual rainfall?

c) Which site has the greater variation in average rainfall?

d) Which site has a greater chance of receiving 300 inches or more of rain?

e) Too much or too little rain affects the water levels in the dam to the point where water skiing is too dangerous. Which site would give a person a better chance of being able to water ski?

8) Consider the following set of data

4, 12, 16, 9, 11, 24, 3, -2, 14, 16, 7, 8, 4, 12, 11, 9, 11, 13, 15, 17, 11, 9, 10

a) Construct a box plot

b) Show by the plot and also by formula that there are two outliers

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9) The following data shows the number of children born to a group of mothers with 6 years or less education, and to mothers with more than 6 years of education

Mother educated for six years or less
14 13 4 14 10 2 13 5 0 0 13 3 9 2 10 11 13 5 14

Mother educated for seven years or more
0 4 0 2 3 3 0 4 7 1 9 4 3 2 3 2 16 6 0 13 6 6 5 9 10 5 4 3 3 5 2 3 5 15 5

a) Draw a box plot for each set of data using the same scale

b) Identify the outliers in either set

c) Comment on the effect of the outliers on the range of the appropriate set, and compare the data without the outliers

d) Compare the medians, IQR, and range with and without the outliers

e) What conclusions can be drawn from the data?

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Exercise 5

Correlation

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1) Plot the following sets of ordered pairs on their own scatter plot

a) (1,2) (2,5) (3,7) (4,8) (5,12) (6,9)

b) (3,4) (6,11) (7,7) (9,30) (11,22) (12,35)

c) (10,12) (9,9) (8,4) (7,8) (6,10) (5,1)

d) (20,8) (14,12) (10,7) (7,10) (3,1) (2,5)

e) (20,2) (10,15) (3,7) (8,4) (5,2) (6,17)

f) (4,12) (2,6) (3,9) (1,3) (5,15) (6,18)

2) For each set of data points in question 1, describe the relationship between the points as strong/medium/weak and positive/negative. Also indicate if any relationship is perfect or there is no relationship at all.

3) For any set of data from question 1 for which there is a relationship, draw the line of best fit through the data, and determine the gradient and vertical intercept. Hence determine the equation of the line of best fit

4) For each of the equations derived in question 3, predict the y value obtained when substituting the point $(3, y)$ into the equation

5) Explain why you could not predict the y value of the point $(40, y)$ in any of the equations above

6) Describe the relation between the two variables of a scatter plot that have the following correlation coefficients

a) $r = 1$

b) $r = 0.8$

c) $r = -0.1$

d) $r = 0.6$

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e) $r = -0.85$

f) $r = 0.09$

7) When the relationship between the sale of blankets in Canada and the sale of air conditioners in Australia at different times of a year is graphed in a scatter plot, the correlation coefficient for the line of best fit is 0.8. Does this mean that the number of air conditioners bought in Australia affects the number of blankets bought in Canada? Explain your answer

8) A scatter plot was produced that showed the relationship between the average life expectancy and the number of television sets per person for a number of countries. The correlation coefficient was very high $(r = 0.92)$. Does this mean that in order to increase life expectancy in third world countries, simply introduce more television sets? Explain your answer

9) Describe the likely scatter plot between the ages and heights of a randomly selected group of 5000 people. What do you think the value of the correlation coefficient may be, and are there any restrictions on the validity of the correlation coefficient? Explain your answer

10) The following table shows the genre of films preferred by different sections of the population (by percentage)

	Action	**Musical**	**Horror**	**Romantic Comedy**
Males 18-25	65	5	15	5
Males 26-50	40	5	30	10
Males 51+	25	30	5	20
Females 18-25	5	15	5	60
Females 26-50	4	26	3	54
Females 51+	3	45	2	40

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a) Why do the percentages not add to 100? Add another column to the table to rectify this

b) Draw a segmented bar chart to represent the data (include your extra column from part a)

11) The back to back stem and leaf plot shows the comparative scores for 2 sports teams

Tigers		Sharks
0 3 7 9	3	2 2
2 8	4	3 5 5
1 3 9 7	5	4 6 8 8 9

Compare the performance of the two teams by use of range, highest and lowest scores and the medians

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Further Mathematics

Number Patterns

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Exercise 1

Arithmetic Sequences

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1) Calculate the value of d in the following sequences

a) 2, 4, 6, 8, 10, ...

b) 1, 4, 7, 10, ...

c) 4, ____, 16, _____, 28

d) 16, 12, 8, 4, ...

e) 64, _____, _____, 28, ...

2) Calculate the value of a in the following sequences

a) ____, 6, 10, 14, ...

b) ____, ____, 15, 18, ...

c) ____, ____, 22, ____, ____, 43

d) ____, ____, ____76, 68, ...

e) ____, ____, ____, 7, ____, ____, ____, ____, 3

3) Find the 5th term of the sequence with first term 4 and a common difference of 3

4) Find the 25th term of the sequence with first term 6 and a common difference of 7

5) Find the common difference of the sequence with a first term of 5 and a twentieth term of 195

6) Find the first term of the arithmetic sequence whose tenth term is 14 and whose twentieth term is 62

7) An arithmetic sequence has a third term of , and a fifteenth term of .

a) What are the values of a and d?

b) List the first three terms of the sequence when

c) List the first 3 terms of the sequence when

8) An arithmetic sequence has a common difference of 4 and a twentieth term of 102. What is the ninth term of this sequence?

9) There are two arithmetic sequences: A and B. $A_{10} = B_{28} = 40$, whilst the value of their first term is the same. If the common difference of sequence A is 3, list the first 4 terms of each sequence

10) Arithmetic sequence A has a first term of (-20) and a twentieth term of 56. Arithmetic sequence B has a first term of 40 and a 5th term of 32. Which term number gives the same value for both sequences, and what is this value?

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Exercise 2

Geometric Sequences

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1) Calculate the value of r in the following sequences

a) 2, 4, 8, 16, ...

b) 3, 4.5, 6.75, ...

c) 20, 10, 5, ...

d) 1000, 200, 40, ...

e) ___, 12, ___, 27, ...

f) ___, ___, 100, ___, 9, ...

2) Calculate the value of a in the following sequences

a) ___, ___, 8, 16, 32

b) ___, ___, 9, ____, 20.25, ...

c) ____, ____, 25, ____, 6.25

d) _____, _____, 100, _____,

6.25

3) Find the 5th term of the sequence with a first term 2 and a common ratio of 3

4) Find the 20th term of the sequence with first term 0.5 and common ratio 4

5) What is the value of the first term of the sequence with an 8th term of 874.8 and a common ratio of 3?

6) A geometric sequence has a first term of -2 and a 10th term of 1024. What is the value of the common ratio?

7) The 2nd term of a geometric sequence is 96 and the 5th term is 1.5. What are the common ratio and the first term?

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8) A geometric sequence has a first term of x and a eleventh term of $59049x$. What is the common ratio of the sequence?

9) The fifth term of a geometric sequence is 48, and the third term is 108. What is the first term and the sixth term?

10) A geometric sequence has a first term of $(x+3)$, and a third term of $\frac{9x+27}{4}$. In terms of x, what is the fifth term?

11) The fifth term of geometric sequence A is 4, and its ninth term is $\frac{1}{4}$. The second term of geometric sequence B is $\frac{1}{16}$, and its fifth term is (-4). Which term number will give the same value for each sequence, and what will this value be?

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Exercise 3

Sum to Infinity

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1) Calculate the sum to infinity of the following sequences

a) $8, 4, 2, 1, \ldots$

b) $10, 1, 0.1, 0.01, \ldots.$

c) 12, 3, 0.75, ...

d) $\frac{5}{8}, \frac{5}{16}, \frac{5}{32}, \ldots.$

e) 6.4, 0.8, 0.1, ...

f) $2, 4, 8, 16, \ldots \ldots$

2) Calculate the following

$$\sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k$$

3) The sum to infinity of a geometric series is 18. If the common ratio is $\frac{2}{3}$, what is the first term of the series?

4) The first term of a geometric series is 21, and its sum to infinity is 28. What is the common ratio?

5) Prove with the use of a geometric series that $0.\dot{9} = 1$

6) Which scenario would get you more money?

- \$10 on day 1 and $\frac{3}{4}$ of what you received the day before from then on
- \$20 on day 1 and $\frac{1}{2}$ of what you received the day before from then on

7) A form of Zeno's paradox (Zeno was a contemporary of Socrates) postulates that one can never walk across a room, since first one must cover half the distance of the room, then half the remaining distance, then half the remaining distance and so on. Since

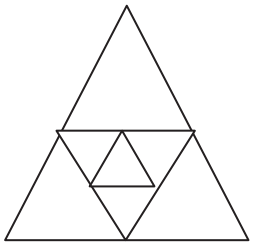
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there will always be a fraction of a distance to cover, the total journey is impossible. Reconcile this paradox with the use of a geometric series

8) A person weighing 210 kg plans to lose 10 kg in the first month of their diet, then 8 kg in the second month, 6.4 kg in the third month, and so on repeating the pattern of weight loss. Their goal is to eventually reach 150 kg. Will they be successful with this strategy? Explain your answer.

9) If the person from question 8 wanted to achieve their goal weight, but maintaining the same pattern of weight loss, how much weight would they have to lose in the first month?

10) An equilateral triangle has a side length of x cm. Another equilateral triangle is inscribed inside the first one such that the vertices of the second triangle sit at the midpoint of the sides of the larger triangle. (See diagram). This process is repeated infinitely. What is the sum of the perimeters of the triangles?



x

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Exercise 4

Difference Equations

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1) Write the first 5 terms of each difference equation

a) $T_{n+1} = T_n + 3, T_1 = 2$

b) $T_{n+1} - T_n = -1, T_1 = 4$

c) $T_n = 1.5T_{n-1}, T_1 = 6$

d) $T_{n+1} \div T_n = 2, T_3 = 10$

e) $T_{n+2} = T_{n+1} - T_n, T_1 = 3, T_2 = 7$

2) Define a difference equation for the following sequences

a) $1, 3, 5, \ldots$

b) $100, 50, 25, 12.5, \ldots$

c) $1, 2, 3, 5, \ldots$

d) $100, 20, 5, \frac{1}{4}$

3) Each day 1000 bacteria are added to a Petrie dish. The bacteria have a 25% chance of survival. Write the linear difference equation that models this situation

4) The temperature of a bath is 12 degrees Celsius. If it is heated by 10% extra every hour, what is the difference equation describing the temperature of the bath each hour, and what will be the temperature in 5 hours from now?

5) You invest $20000 at 24% p.a. interest compounded monthly.

a) Model the above using a difference equation

b) Calculate the value of your investment after 3 months

c) Produce an equation for the value of your investment after n months

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6) The dose of a drug is 700 mg. If the drug dissipates at the rate of 10% of the amount of drug left per hour, how long will it take for the level of the drug to fall below 300 mg?

7) The population of a school of fish was 300 in the year 2010, and the school grows by 5% per year.

a) Write a difference equation for the population of the school

b) When will the population reach 400?

8) You are learning vocabulary, and each day you memorise 25 words, but also forget 10% of all the words you have previously learned.

a) Write the difference equation that models this situation

b) How many words will you have memorised after n days?

c) How many words can you ever hope to learn?

9) If F_n is the n_{th} term of the Fibonacci Sequence, show that

a) $F_{n+1} \times F_{n-1} - F_n^2 = (-1)^n$

b) $F_{n+k} = F_k \times F_{n+1} + F_{k-1} \times F_n$

10) A cow weighs 100 kg and gains 2 kg per day with its food costing 50 cents per day. The price for cows is currently $1 per kg, but is falling by 5 cents per day. In how many days should the cow be sold to maximize profit?

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Further Mathematics

Geometry & Trigonometry

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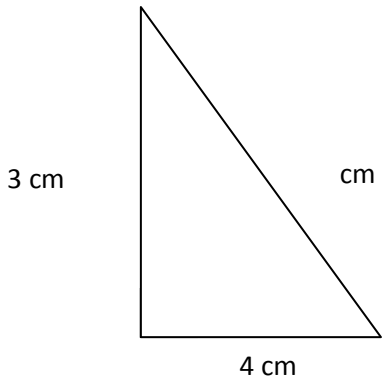
Exercise 1

Pythagoras' Theorem

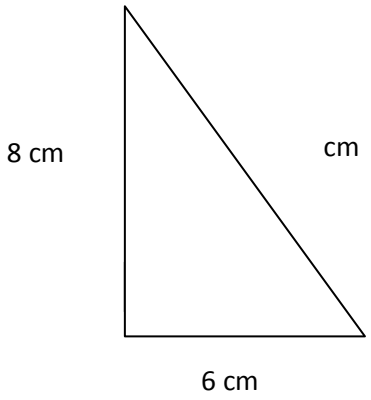
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1) Find the value of to 2 decimal places in the following diagrams

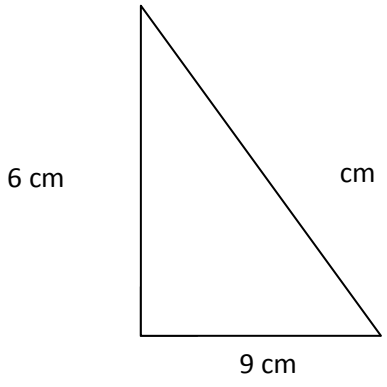
a)



b)

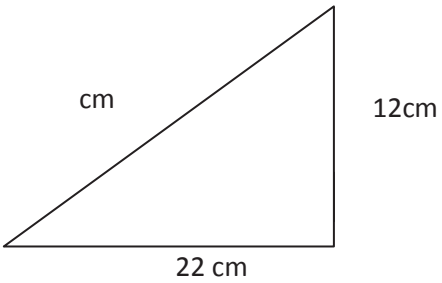


c)

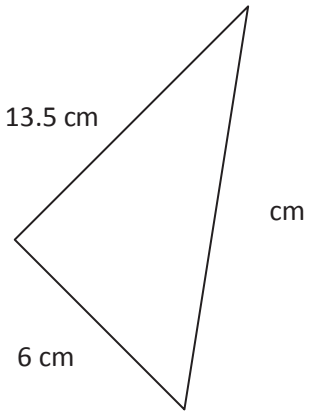


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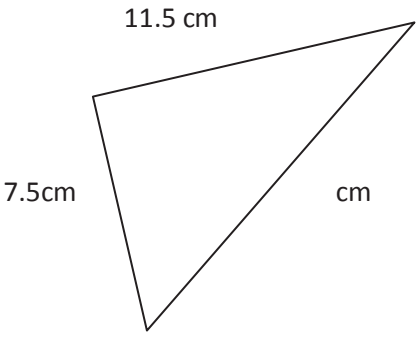
d)



e)



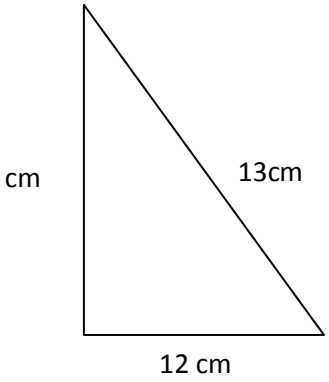
f)



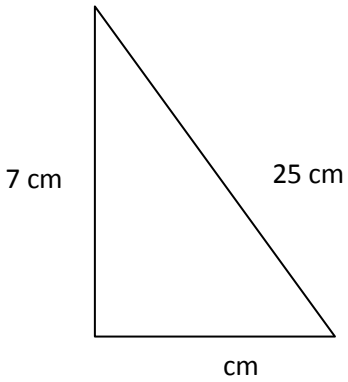
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2) Find the value of to 2 decimal places in the following diagrams

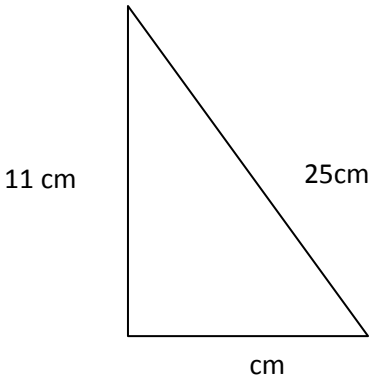
a)



b)

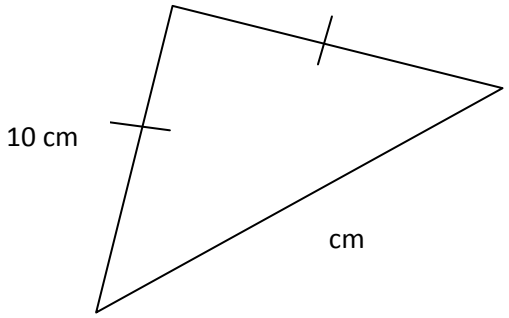


c)

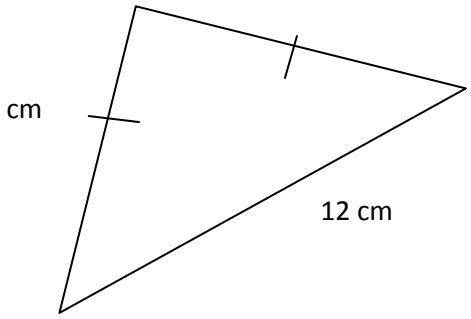


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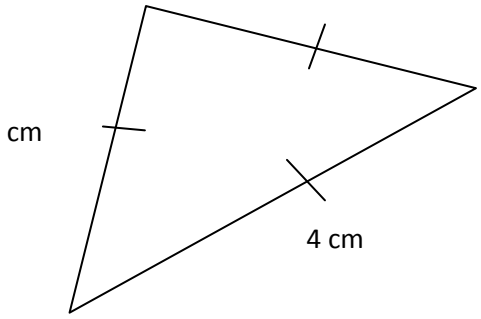
d)



e)



f)



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3) A man walks 5 km east then turns and walks 8 km south. How far is the shortest distance to his starting position?

4) A ladder 2 meters long is placed against a wall and reaches 1.5 meters up the wall. How far is the foot of the ladder from the base of the wall?

5) A farmer wishes to place a brace across the diagonal of a rectangular gate that is 1.8 metres long and 0.6 metres wide. How long will the brace be?

6) A square room measures 11.7 metres from corner to corner. How wide is it?

7) The size of television sets are stated in terms of the diagonal distance across the screen. If the screen of a set is 40 cm long and 30 cm wide, how should it be advertised?

8) A student has two choices when walking to school. From point A, he can walk 400 metres, then turn 90° and walk a further 200 metres to point B (school), or he can walk across the field that runs directly from A to B. How much further does he have to walk if he takes the path instead of the field?

9) A rod is to be placed in the box below, and it will only fit in from the top left to the bottom right of the box. What is the length of the rod?



10) A pyramid is made up of a square and four equilateral triangles of side length 10 cm. What is the vertical height of the pyramid?

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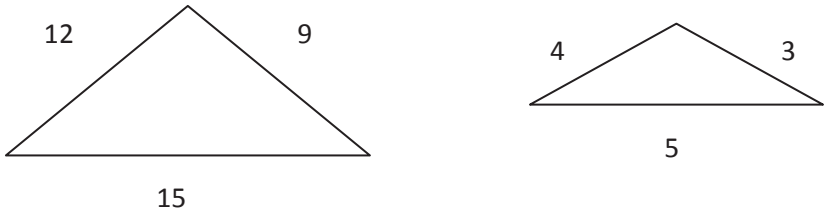
Exercise 2

Similarity

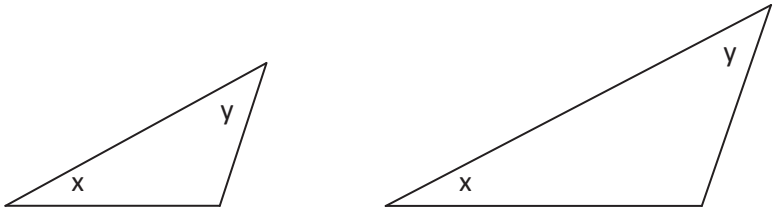
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1) Decide if the following triangles are similar, and if so state the similarity conditions

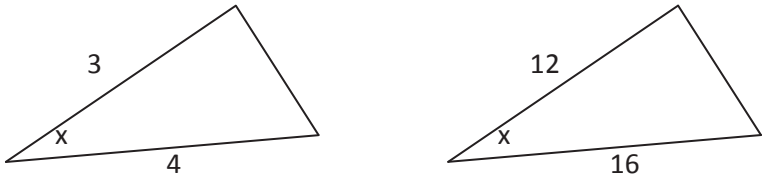
a)



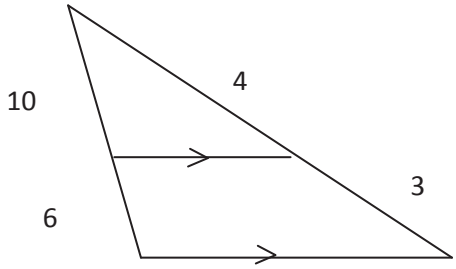
b)



c)

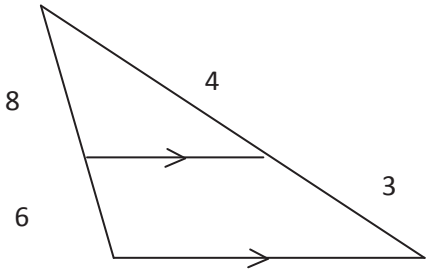


d)

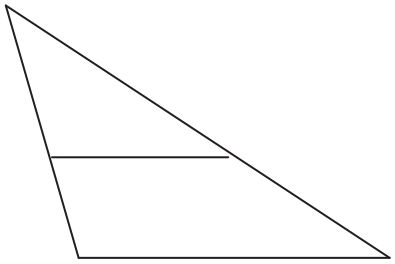


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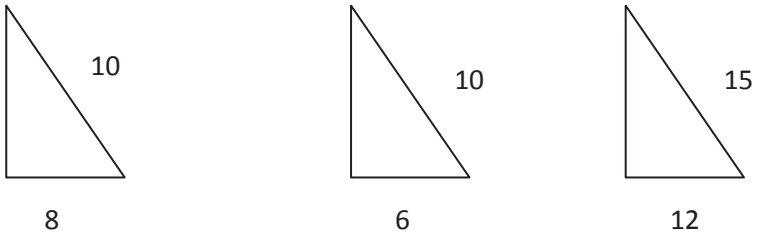
e)



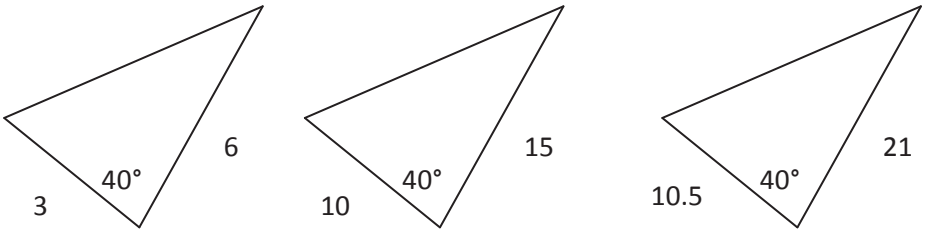
2) What additional information is needed to show that the two triangles are similar by AAA?



3) Of the following three right-angled triangles, which two are similar and why?

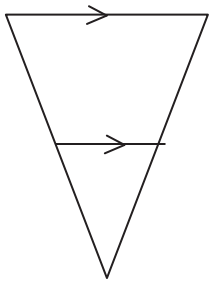


4) Of the following three triangles, which are similar and why?



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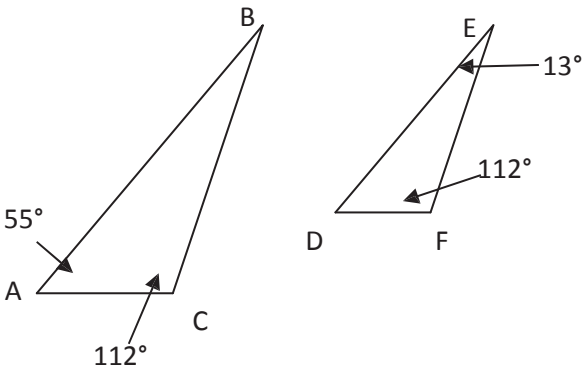
5) Prove that the two triangles in the diagram are similar



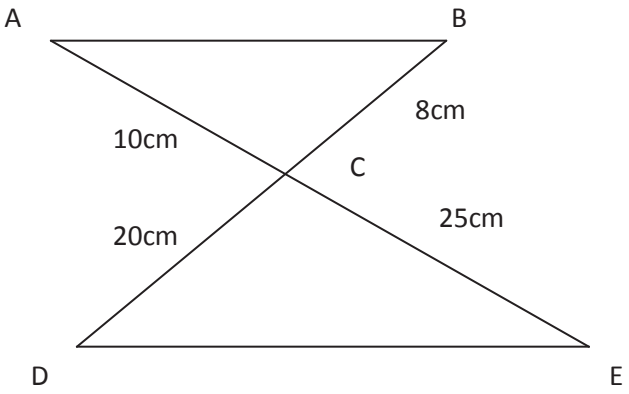
6) Prove that if two angles of a triangle are equal then the sides opposite those angles are equal

7) Determine if each pair of triangles is similar. If so, state the similarity conditions met

a)



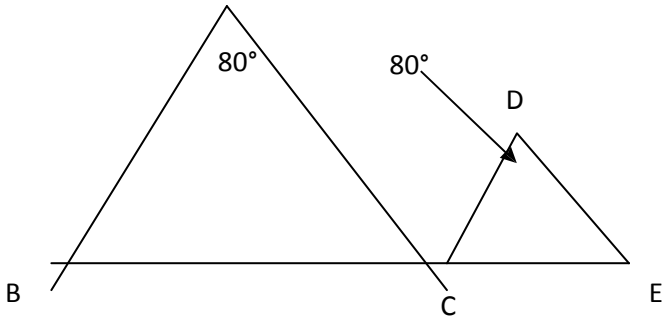
b)



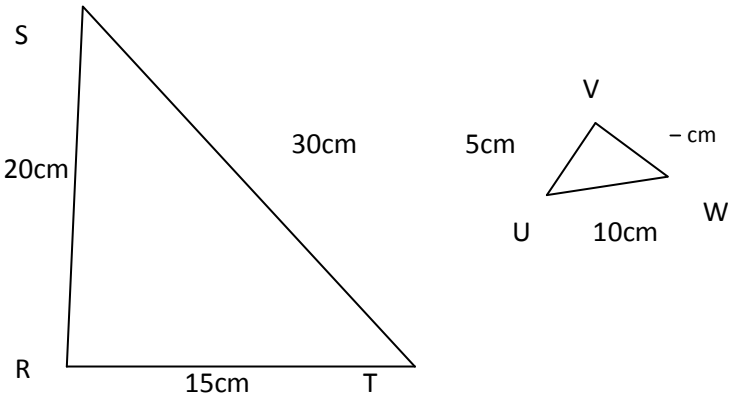
A

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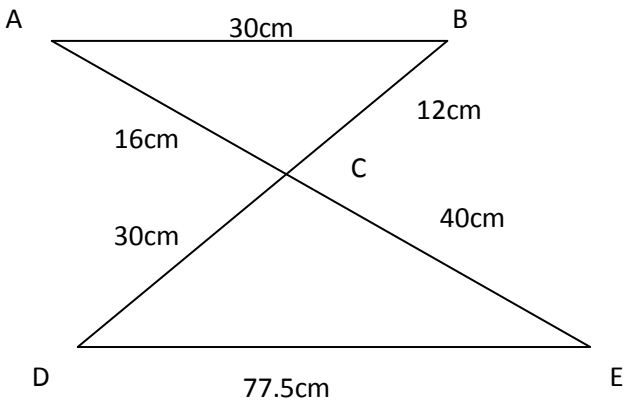
c) AB || DC



d)

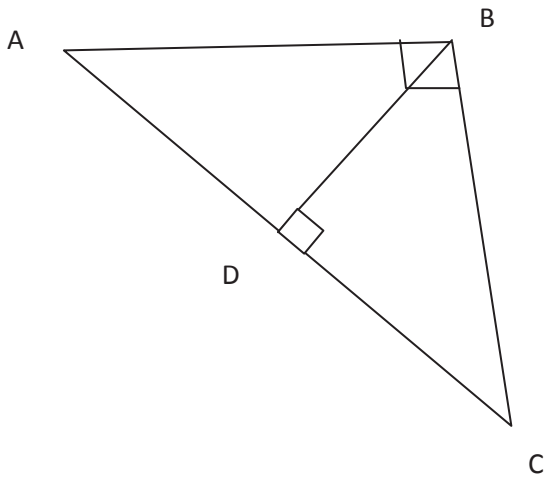


e)



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f)



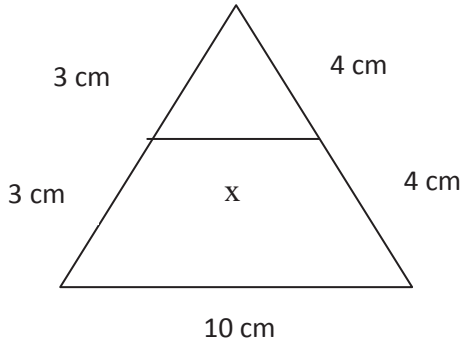
8) A tower casts a shadow of 40 metres, whilst a 4 metre pole nearby casts a shadow of 32 metres. How tall is the tower?

9) A pole casts a 4 metre shadow, whilst a man standing near the pole casts a shadow of 0.5 metres. If the man is 2 metres tall, how tall is the pole?

10) A ladder of length 1.2 metres reaches 4 metres up a wall when placed on a safe angle on the ground. How long should a ladder be if it needs to reach 10 metres up the wall, and be placed on the same safe angle?

11) A man stands 2.5 metres away from a camera lens, and the film is 1.25 centimetres from the lens (the film is behind the lens). If the man is 2 metres tall how tall is his image on the film?

12) What is the value of in the following diagram?



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Exercise 3

Volume & Surface Area

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1) Calculate the surface area of the following right closed cylinders

a) Radius of 2mm, and a vertical height of 10mm

b) Radius of 1cm and a vertical height of 5mm

c) Vertical height of 500mm and a radius of 0.3m

d) Vertical height of 2.5m and a radius of 2000mm

2) Calculate the surface area of the following

a) A sphere having a radius of 10cm

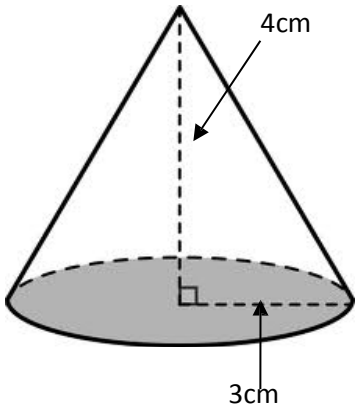
b) A sphere having a diameter of 350mm

c) A hemisphere having a radius of 8m

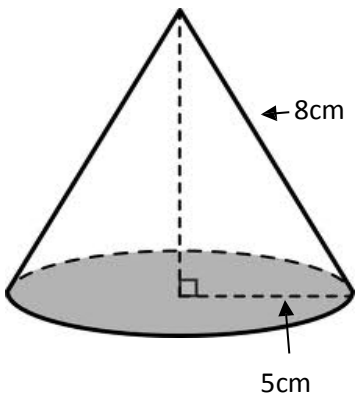
d) A hemisphere having a diameter of 16mm

3) Calculate the surface area of the following.

a)

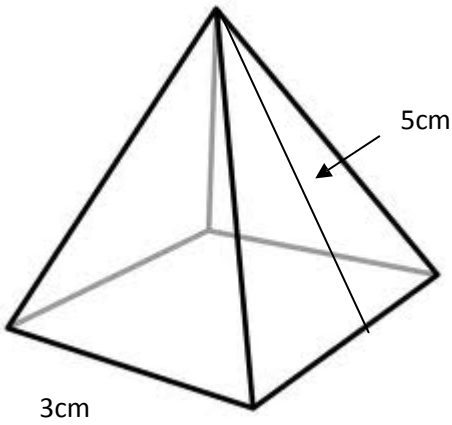


b)

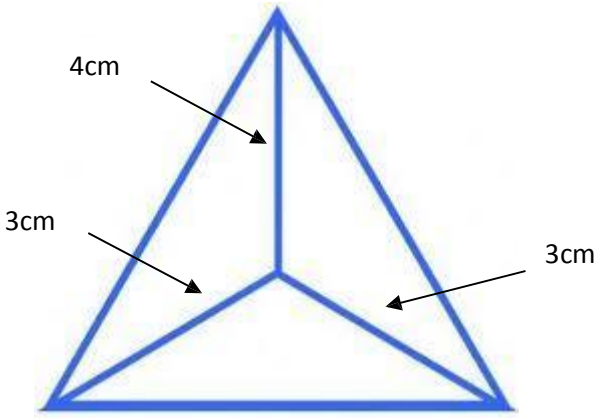


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c)



d)



4) Calculate the volume of a right cone of radius 2cm and a vertical height of 5cm

5) Calculate the volume of a square pyramid of side length 2cm and vertical height 5cm.

6) A cone has a volume of 30π cm^3. If the side length of a square pyramid is 6cm, what must its height be in order to have the same volume as the cone?

7) Calculate the volume of the following

a) A sphere having a radius of 2mm

b) A sphere having a diameter of 20cm

c) A hemisphere having a radius of 3.2m

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d) A hemisphere having a diameter of 40mm

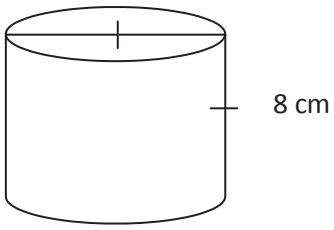
8) The volume of a sphere is $\frac{256}{3}\pi$ mm^3. What is its radius?

9) A square base pyramid has a perpendicular height of 10mm. What is the length of each side of the base if its volume is 270 mm^3?

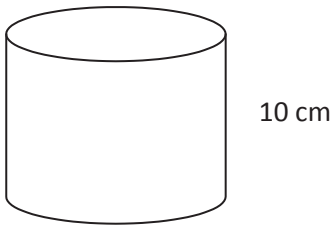
10) A sphere of volume $\frac{4}{3}\pi$ m^3 fits exactly inside a cube. What is the volume of the cube?

11) Calculate the surface area of the following cylinders (parts c and d are open cylinders; they have no top or bottom)

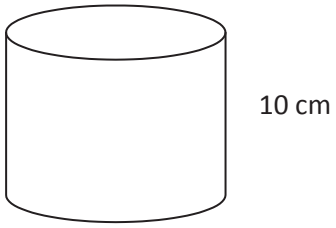
a)



b)

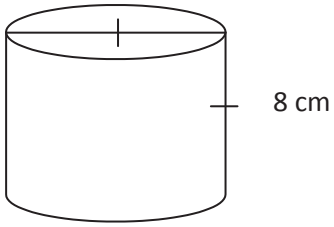


c)

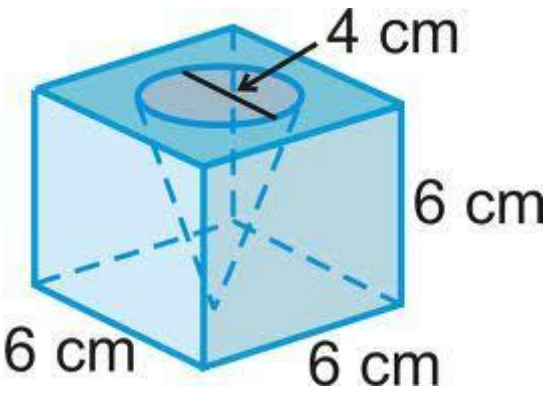


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d)

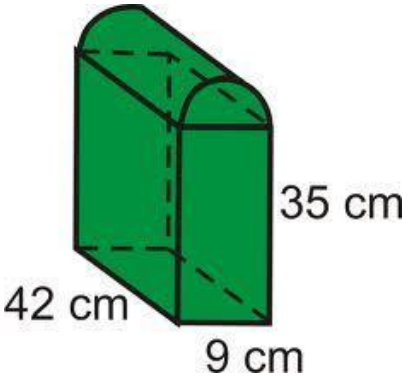


12) What is the total surface area of the following solid, which is a cube with a conic section cut out?



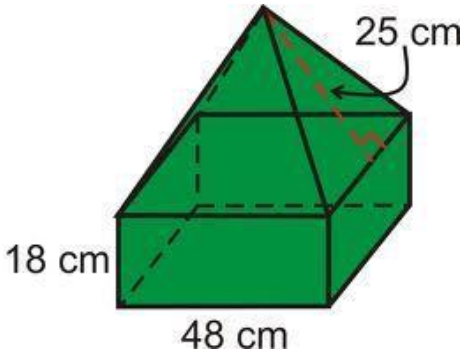
13) Calculate the volume of the following solids

a)

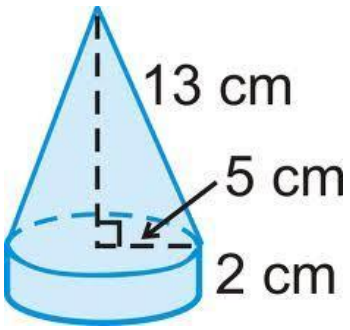


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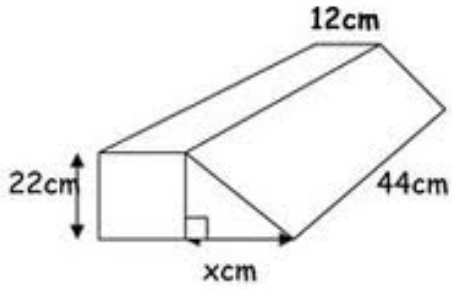
b)



c)



14) The volume of the solid below is 16456 cm^3. What is the value of x?



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Exercise 4

Change of Scale

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1) An equilateral triangle has a perimeter of 9 cm. If the lengths of the sides are scaled by a factor of 2.5, what is the new perimeter of the triangle?

2) The sides of an equilateral triangle are increased by a factor of 3.

a) What is the effect on the perimeter of the triangle?

b) What is the effect on the area of the triangle?

3) The sides of a square are scaled by a factor of ½.

a) What effect does this have on the perimeter of the square?

b) What effect does this have on the area of the square?

4) The length of each side of a cube is tripled

a) What effect does this have on the surface area of the cube?

b) What effect does this have on the volume of the cube?

5) A triangular prism has an isosceles right angled triangle as its base. If all sides Are doubled:

a) What effect will this have on the surface area of the prism?

b) What effect will this have on the volume of the prism?

6) The radius of a circle is scaled by a factor of 4; what effect will this have on the area of the circle?

7) The diameter of a cylinder is doubled and its height tripled.

a) What effect will this have on the surface area of the cylinder?

b) What effect will this have on the volume of the cylinder?

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8) A projector magnifies a picture in front of it onto a large screen. The image is in the shape of a rectangle with length twice its width. The area of the image on the screen is 7.2 square metres. If the projector magnifies the area of the original picture by a factor of 10, what are the dimensions of the picture?

9) A company manufactures globes of the world. To make a prototype it used of paper. If the **volume** of the actual globe is twice that of the prototype, how much extra paper will they use?

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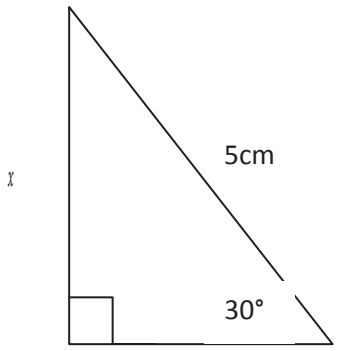
Exercise 5

Trigonometry (I)

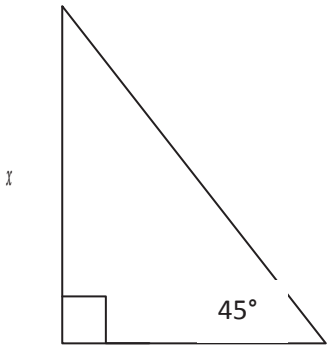
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1) Calculate the length of x in each of the diagrams below

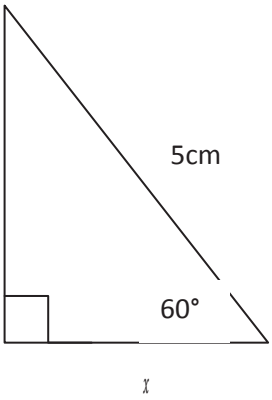
a)



b)

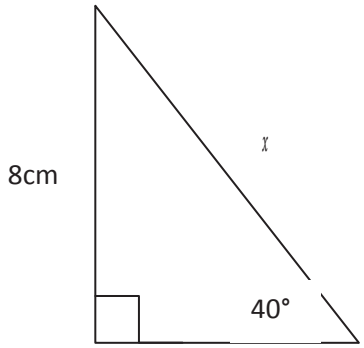


c)



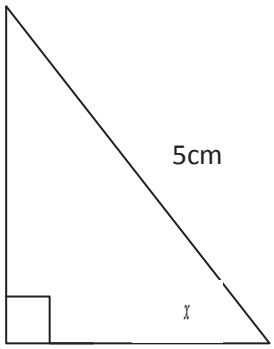
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d)



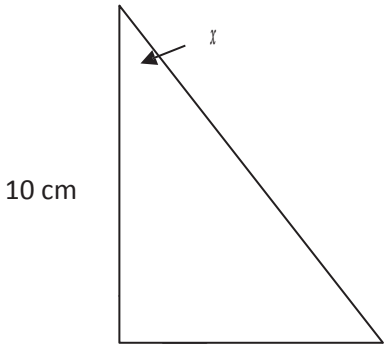
2) Calculate the size of angle x in the diagrams below, correct to the nearest degree.

a)



3 cm

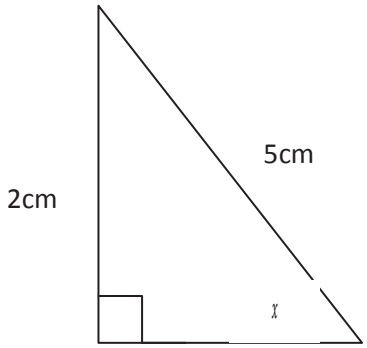
b)



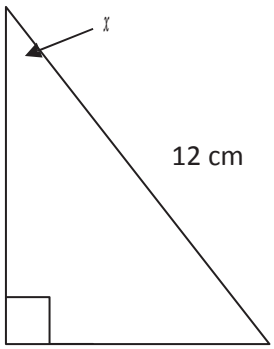
6cm

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c)



d)

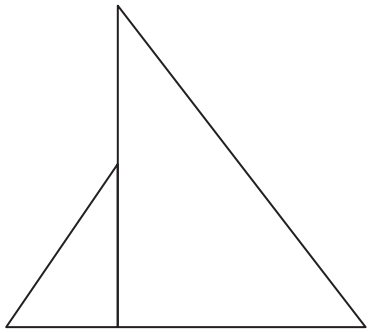


6 cm

3) The foot of a ladder is 3 metres away from the base of a wall. If the ladder reaches 4.5 metres up the wall, what angle does the foot of the ladder make with the ground?

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4) Two sails sit back to back on a yacht. The first sail reaches half way up the second

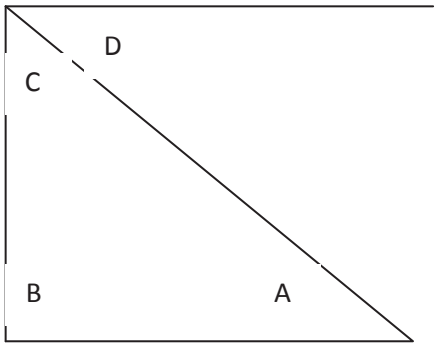


The longest part of the second sail is 4 metres, and it makes an angle of 50 degrees to the deck. If the longest part of the first sail is 3 metres, what angle does it make with the deck?

5) A piece of carpet is in the shape of a right angled triangle. The longest side is 80 cm, and it makes an angle of 65 degrees with the next side. What is the area of the piece of carpet?

6) Tom walks at an average speed of 4 km per hour in a north east direction. Ben walks at 5 km per hour, starting from the same point but in a south east direction. After 3 hours what is the shortest distance between them, and what is the angle from Tom to Ben?

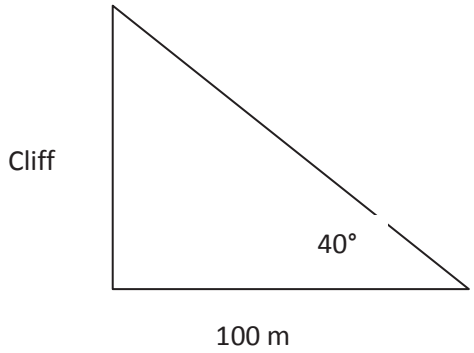
7) Identify the angles of elevation and depression in the diagram below



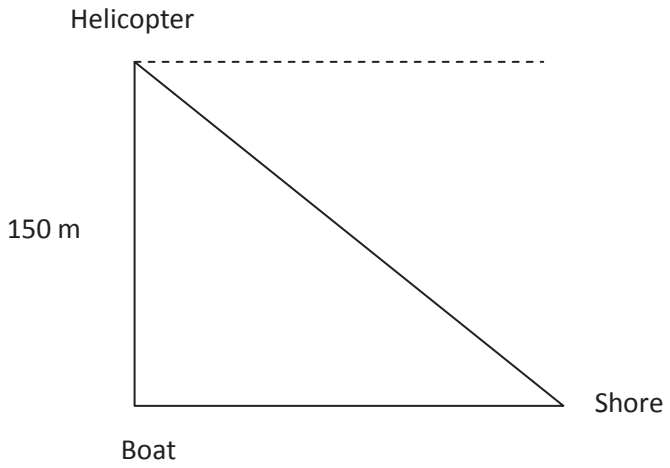
Complete the statement: The angle of elevation is ..... the angle of depression

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8) A man standing 100 metres away from the base of a cliff measures the angle of elevation to the top of the cliff to be 40 degrees. How high is the cliff?



9) A helicopter is hovering 150 metres above a boat in the ocean. From the helicopter, the angle of depression to the shore is measured to be 25 degrees. How far out to sea is the boat? (You need to fill in angle of depression on diagram)



10) A ramp is built to allow wheelchair access to a lift. If the angle of elevation to the lift is 2 degrees, and the bottom of the lift is 50 cm above the ground how long is the ramp?

11) The angle of elevation to the top of a tree is 15 degrees. If the tree is 10 metres tall how far away from the base of the tree is the observer?

12) From the top of a tower a man sees his friend on the ground at an angle of depression of 30 degrees. If his friend is 80 metres from the base of the tower how tall is the tower?

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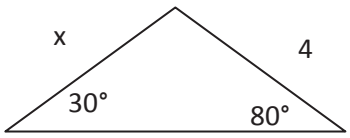
Exercise 6

Trigonometry (II)

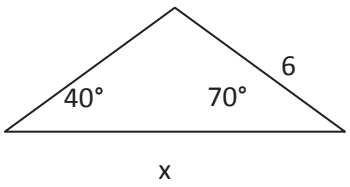
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1) Solve the following using the sine rule. Note for questions where the angle is unknown, round your answer to one decimal place, and ensure all possible solutions are found. (Diagrams are not drawn to scale)

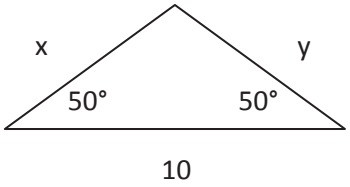
a) a



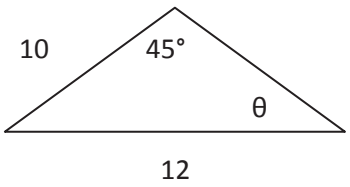
b)



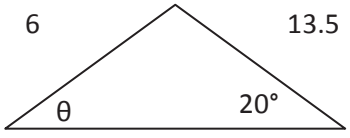
c)



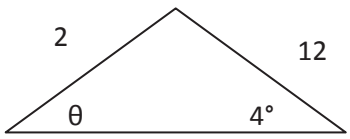
d)



e)



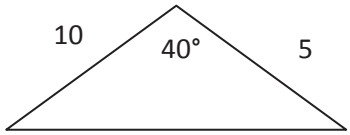
f)



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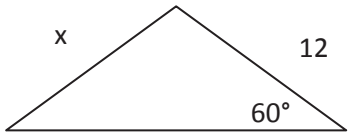
2) Solve the following using the cosine rule. Note for questions where the angle is unknown, round your answer to one decimal place, and ensure all possible solutions are found. (Diagrams are not drawn to scale)

a)



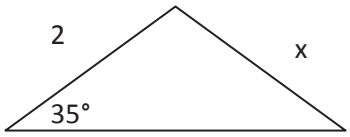
x

b)



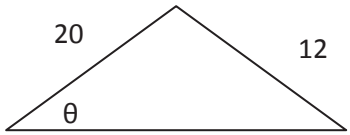
13

c)



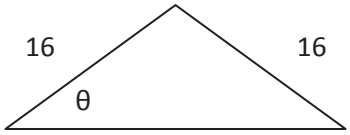
30

d)



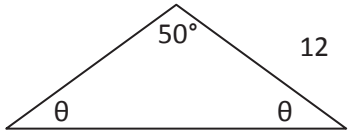
25

e)



24

f)



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3) Find the area of the triangles in question 2 by using the sine formula

4) Solve the following by using the sine rule or cosine rule; draw a diagram to help solve

a) A post has been hit by a truck and is leaning so it makes an angle of 85° with the ground. A surveyor walks 20 metres from the base of the pole and measures the angle of elevation to the top as 40°. How tall is the pole if it is leaning toward him? How tall is the pole if it is leaning away from him?

b) Boat A travels due east for 6 km. Boat B travels on a bearing of 130° for 8 km. How far apart are the boats?

c) A mark is made on the side of a wall. A man 40 metres from the base of the wall measures the angle of elevation to the mark as 20°, and the angle of elevation to the top of the wall as 60°. How far is the mark from the top of the wall?

d) What is the perimeter of a triangle with two adjacent sides that measure 15 and 18 metres respectively, with the angle between them 75°?

e) The pilot of a helicopter hovering above the ocean measures the angle of depression to ship A on its left at 50°, and the angle of depression to ship B on its right at 70°. If the ships are 200 metres apart, how high above the ocean is the helicopter hovering?

f) A car travels 40 km on a bearing of 70°; then travels on a bearing of 130° until it is exactly due east of its starting position . What is the shortest distance back to its starting position?

5) Find the areas of the triangles used in question 4 parts a, b and d

6) A point K is 12km due west of a second point L and 25km due south of a third point M. Calculate the bearing of L from M

7) Point Y is 1km due north of point X. The bearings of point Z from X and Y are 26° and 42° respectively. Calculate the distance from point Y to point Z.

8) A ship steams 4km due north of a point then 3km on a bearing of 040°. Calculate the direct distance between the starting and finishing points.

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9) The bearings of a point Z from two points X and Y are 30° and 120°. The distance from X to Z is 220km. What is the distance of X from Y?

10) A man walked along a road for 6km on a bearing of 115°. He then changed course to a bearing of 25° and walked a further 4km. Find the distance and bearing from his starting point

11) Directly east of a lookout station, there is a small forest fire. The bearing of this fire from another station 12.5 km. south of the first is 57°. How far is the fire from the southerly lookout station?

12) Mark and Ron leave a hostel at the same time. Mark walks on a bearing of 050° at a speed of 4.5 kilometres per hour. Ron walks on a bearing of 110° at a speed of 5 kilometres per hour. If both walk at steady speeds, how far apart will they be after 2 hours??

13) A ship leaves a harbour on a bearing of 50° and sails 50km. It then turns on a bearing of 120° and sails for another 40km. How far is the ship from its starting point?

14) Two ships A and B are anchored at sea. B is 75km due east of A. A lighthouse is positioned on a bearing of 045° from A and on a bearing of 320° from B. Calculate how far the lighthouse is from the ships

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Further Mathematics

Graphs & Relations

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Exercise 1

Linear Relationships

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1) When baking scones the oven must be set at 150 degrees Celsius plus 2 degrees extra per scone

Draw a table that shows what temperature an oven must be on to cook 10, 20, 30, 40 and 50 scones and graph the relationship using appropriate scale

Can the points on the graph be joined up to form a line? Why or why not?

2) A river has a stepping stone every 1.5 metres. Draw a table showing the relationship between the number of stones and the distance travelled across the river. Draw a graph that shows the relationship. Explain why the points should not be joined to form a line

3) A boy places three lollies into a jar. Every minute he puts in another lolly.

a) Draw a table that shows how many lollies in the jar after each minute

b) Graph the relationship

c) Explain why the points should not be joined

4) Alan has 20 CDs in his collection. At the end of each month he buys a CD

a) Draw a table that shows how many CDs in his collection each month

b) Graph the relationship

c) Explain why the points should not be joined

5) The instructions for cooking a roast state that it should be cooked for thirty minutes plus 40 minutes for every kg the meat weighs

a) For how long should a roast that weighs 1.5 kg be cooked for?

b) Construct a table of values that relate the weight of the meat to its cooking time

c) Graph the values

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d) Determine the gradient of the line produced. How does this value relate to the quantities in the problem?

e) Relate the y intercept to the quantities in the problem

f) Is the graph valid for all weights; that is can the graph be extended indefinitely? Explain your answer

6) A plumber charges a call out fee of $25 plus $20 per hour for his work. If he works for part of the hour he only charges for that part. For example, for 15 minutes work he will charge $5 (plus his call out fee)

a) How much will he charge for 2 hours work?

b) How much will he charge for 3.5 hours work

c) Construct a table of values that relate the time taken for a job to the total charge

d) Graph the values

e) Determine the gradient of the line produced. How does this value relate to the quantities in the problem

f) Relate the y intercept to the quantities in the problem

g) Is the graph valid for all times; that is can the graph be extended indefinitely? Explain your answer

7) Another plumber charges a $25 call out fee and $20 per hour for his work. Differently to the previous plumber he charges $20 even if he only works for part of an hour. For example, for 15 minutes work he will charge $20 (plus his call out fee)

a) How much will he charge for 2 hours work?

b) How much will he charge for 3.5 hours work

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c) Construct a table of values that relate the time taken for a job to the total charge

d) Graph the values

e) How does the graph differ from that in question 6?

8) To convert from Celsius to Fahrenheit temperature the following formula is used

$$F = \frac{9}{5}C + 32$$

a) Construct a table of values for $C = 0 \text{ to } 40$ in steps of 5 degrees

b) Graph the relationship

c) Determine the gradient of the line produced. How does this value relate to the quantities in the equation?

d) Relate the y intercept to the quantities in the equation

e) Use the graph to extrapolate the value of 42 degrees Celsius in Fahrenheit

f) Use the graph to determine how many degrees Celsius equals 23 degrees Fahrenheit

g) Is the graph valid for all values of C? Explain

9) One Australian dollar currently buys 56.5 Indian rupees

a) Construct a table of values for 0 to 30 Australian dollars in steps of 5 dollars

b) Graph the relationship

c) Determine the gradient of the line produced. How does this value relate to the quantities in the equation?

d) Relate the y intercept to the quantities in the equation

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e) How many rupees does 40 Australian dollars buy?

f) How many Australian dollars does 1695 rupees buy?

10) A bath has 200 litres of water in it. The plug is pulled and water flows from it at the rate of 4 litres per second.

a) Construct a table of values that relate the volume of water in the bath to the time since the plug was pulled

b) Graph the relationship

c) From your graph how long until the bath is empty?

d) Determine the gradient of the line produced. How does this value relate to the quantities in the problem?

e) Relate the y intercept to the quantities in the problem

f) Is the graph valid for all values of t? Explain

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Exercise 2

Simultaneous Equations

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1) Solve the following simultaneous equations by using the guess check and improve method

a) $2x + y = 4 \text{ and } x - y = 5$

b) $x + y = 8 \text{ and } 2x - y = 10$

c) $2x + y = 3 \text{ and } 3x - 2y = 1$

d) $x - 3y = -4 \text{ and } x + y = 4$

e) $x + 2y = 5 \text{ and } x - 2y = 1$

f) $x + 3y = 7 \text{ and } x - y = 3$

g) $2x - y = -1 \text{ and } 5x + 7y = 7$

2) For each of the simultaneous equations in question 1, make a table of possible values and use it to check each of your solutions

3) Graph each pair of simultaneous equations from question 1, and use your graphs to check each of your solutions

4) Use an algebraic method (substitution, subtraction or addition of equations) to solve each pair of simultaneous equations from question 1.

5) Solve the following word problems by generating a pair of simultaneous equations and solving them by any of the methods used above. Check your solutions by substituting back into the original equations

a) The sum of two numbers is 8 and the difference is 4. Find the numbers.

b) The cost of two rulers and a pen is $6.00. The difference of cost between 3 rulers and 2 pens is $2.00. Find the cost of a ruler and a pen.

c) If I double two numbers and then add them together I get a total of 8. If I multiply the first number by 3, then subtract the second number I get 4. What are the two numbers?

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d) The average of two numbers is 9. The difference is 6. Find the numbers

e) There are two angles on a straight line. One angle is 45 more than twice the other. Find the size of each angle.

f) The length of a rectangle is twice its width. The perimeter is 42. Find its dimensions

6) One thousand tickets to a show were sold. Adult tickets cost $8.50 and children's were $4.50. $7300 was raised from the sale of the tickets. How many of each type were sold?

7) Mrs. Brown. invested $30,000; part at 5%, and part at 8%. The total interest on the investment was $2,100. How much did she invest at each rate?

8) Tyler is catering a banquet for 250 people. Each person will be served either a chicken dish that costs $5 each or a beef dish that costs $7 each. Tyler spent $1500. How many dishes of each type did Tyler serve?

9) Your teacher is giving you a test worth 100 points containing 40 questions. There are two-point and four-point questions on the test. How many of each type of question are on the test?

10) The cost to hire a hall for a lecture is $500 plus insurance of $10 per person who attends. The organisers are getting a subsidy of $50 and they are charging each attendee $25. How many people must attend in order for the organisers to break even?

11) Blaxland ceramics manufactures discs for power poles. They buy each disc for $10, machine them and sell them for $22 each. They pay $4000 a week in wages and $2000 a week in rent and overheads. They receive $60 per week in wage subsidies. How many ceramic discs must they sell in a week to break even?

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Exercise 3

Non-linear Relationships

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1) Graph each quadratic equation below, by first making a table of values

a) $y = x^2$

b) $y = 2x^2$

c) $y = 4x^2$

d) $y = 0.5x^2$

e) $y = \frac{1}{2}x^2$

f) $y = -2x^2$

2) Graph each quadratic equation below, by first making a table of values

a) $y = x^2 + 1$

b) $y = x^2 + 3$

c) $y = x^2 - 2$

d) $y = x^2 - \frac{1}{2}$

e) $y = 2x^2 - 3$

3) Graph the following

a) $y = (x + 1)^2$

b) $y = (x + 2)^2$

c) $y = (x - 2)^2$

d) $y = (x - 1)^2$

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4) Graph each parabolic equation below, by first making a table of values

a) $y = \frac{1}{x}$

b) $y = \frac{3}{x}$

c) $y = \frac{5}{x}$

d) $y = \frac{1/2}{x}$

e) $y = \frac{-2}{x}$

5) Identify which of the following equations produce lines, parabolas or hyperbolae when graphed

a) $y = \frac{2}{x} + 4$

b) $y = 3x - 6$

c) $y = \frac{1/3}{x}$

d) $y = \frac{1}{2}x^2 + 1$

e) $y = \frac{1}{2}x$

f) $y = -\frac{1}{2x} + 2$

g) $y = -2 - \frac{1}{4}x^2$

6) Graph the following equations

a) $y = x^3$

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b) $y = x^3 + 1$

c) $y = x^3 - 1$

d) $y = x^3 + 2$

e) $y = x^3 - 3$

7) Graph the following equations

a) $y = 2^x$

b) $y = 3^x$

c) $y = \left(\frac{1}{2}\right)^x$

d) $y = 5^x$

8) Graph the following equations

a) $y = 2^{x+1}$

b) $y = 2^{x-1}$

c) $y = 2^{x+2}$

d) $y = 2^{x-3}$

9) The makers of a part for lawnmowers analysed profitability for various levels of production and discovered that the profitability was modelled by the equation

$P = 20n - n^2$

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Where n is the number of units produced and P is the total profit

a) Graph the equation

b) For how many units is the total profit zero?

c) How many units of production generate the maximum profit, and what is this profit?

d) The company wishes to make a profit of $75. How many units should it produce?

10) A company can sell the part it makes for varying prices depending on the quantity produced. The equation that relates the quantity produced to the price is

$$p = 70 - q$$

The company has fixed costs of $500, and it costs them $10 to make each unit

Graph the profit function for the company, and use it to determine the level of production needed to break even, and the point at which maximum profit is made.

HINT: Profit = (price x quantity)-total costs

11) The population of a strain of bacteria is modelled by the equation $n = 2x^2$ where x is the number of hours since the colony was begun. The number of bacteria that the food supply can sustain is described by the equation $n = 1024x^{-1}$

a) For how many hours is the population sustainable?

b) What is the maximum sustainable population?

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Exercise 4

Proportional Relationships

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1) Graph the following relations

a) $y = x^{-2}$

b) $y = x^{-1}$

c) $y = x$

d) $y = x^2$

e) $y = x^3$

2) Graph the following

a) $y = 2x^{-2}$

b) $y = 3x^{-2}$

c) $y = 2x^3$

d) $y = 4x^3$

3) Comment on the effect of changing the value of k and n in equations of the form $y = kx^n$

4) The following table shows values for an equation of the form $y = kx^2$.

x	y
1	2
2	8
3	18
4	32

a) Graph y against x^2 and find the gradient of the line

b) Hence determine the value of k, in the original equation

5) The following table shows values for an equation of the form $y = kx^3$

x	y
2	12
4	96
5	187.5
8	768

a) Graph y against x^3 and find the gradient of the line

b) Hence determine the value of k in the original equation

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Exercise 5

Linear Programming

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1) Shade the region(s) of the number plane defined as follows

a) The region where $y < 1 - x$ and $2y > x + 2$

b) The region where $y > x + 2$ and $-\frac{1}{2}y > \frac{1}{2}x + 4$

c) The region where $2x + y < 4$ and $x - 2y < -3$

d) The region where $(-x - y) > 0$ and $\frac{1}{2}x < -(y + 1)$

e) The region bounded by the inequalities $2y < 3x - 1$, $-y > 2x - 10$ and $3y > x + 2$

2) Write the following as linear equations

a) The number of pencils produced per hour cannot exceed 20

b) It costs \$3 to make each peg

c) The total cost of manufacturing car doors per day is \$400 fixed plus \$25 per door made

d) The total revenue for a day's production of x widgets that sell for \$8 each and y wodgets that sells for \$10 each.

e) The wage budget for a factory is \$4000 per day. They employ some tradesmen at \$200 per day and some assistants at \$125 per day

3) For each part, graph the 3 equations and determine the points of intersection

a) $y = -x + 4$

$y = 3x - 2$

$y = 1$

b) $y = \frac{1}{2}x + 1$

$y = -3x + 6$

$y = 2x + 1$

c) $y = 3x$

$y = -\frac{1}{2}x + 6$

$y = x$

4) The Paint Barn sells two blends of paint in 4 litre tins. Blend A contains one quarter yellow and

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three quarters blue paint. Blend B contains half of each colour.

Profit on blend A is $4 per can, and on blend B is $5 per can.

Each week the store has 100 litres of yellow and 200 litres of blue paint available for mixing. All of the paint should be used in the blends.

How many of each blend should be made in order to maximize profit and what is the maximum profit?

5) A firm manufactures two types of tiles; plastic and ceramic

At least 2 boxes of ceramic tiles must be made in one day, and the factory must produce at least 10 boxes in total.

It takes 1 hour to make a box of plastics and half an hour to make a box of ceramics. The factory operates for 16 hours per day.

a) If the profit was $100 per box of plastics, and $400 per box of ceramics, what production would give the maximum profit?

b) What production would give the maximum profit if the profit on each type was $200 per box?

6) A senior citizens group is preparing a trip for 400 residents. The company who is providing the transportation has 10 buses of 50 seats each and 8 buses of 40 seats, but only has 9 drivers available. The rental cost for a large bus is $800 and $600 for the small bus.

Calculate how many buses of each type should be used for the trip for the least possible cost, and what is that cost?

7) Coopers Heating manufactures bar heaters and air heaters. The manufacturing plant has the capacity to manufacture at most 600 bar heaters and 500 blowers. It costs the company $10 to make a bar heater and $12 to make a blower. The company can spend $8400 to make these products. Coopers Heating makes a profit of $19 on each bar heater and $12 on each blower. To maximize profits, how many of each product should they manufacture?

8) Cook Island Cruises sells “A” class and “B” class seats for its day tour. To charter a boat at least 5 “A” class tickets must be sold and at least 9 “B” class tickets must be sold. The boat does not hold more than 30 passengers. The company makes $40 profit for each “A” class ticket sold and $45 profit for each “B” class ticket sold. In order for Cook Island Cruises to maximize its

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profits, how many "B" class seats should they sell?

9) The Waggles, a world famous children's entertainment group will appear at the Melbourne Cricket Ground. According to MCG and safety policy, no more than 2000 adult tickets can be sold and no more than 4000 children's tickets can be sold. It costs $0.50 per ticket to advertise the band to children and $1 per ticket to advertise to adults. The group has an advertising budget of $3000 for this show. Find the maximum profit the company can make if it charges $4 for a child's ticket and $7 for an adult ticket. How many children's tickets should they sell?

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Further Mathematics

Networks

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Exercise 1

Representation of Networks

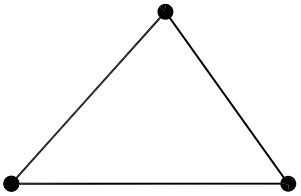
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1) For each of the graphs below, name and list each vertex, and whether it is a vertex of odd or even degree

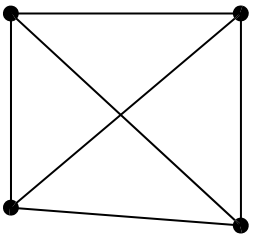
a)



b)

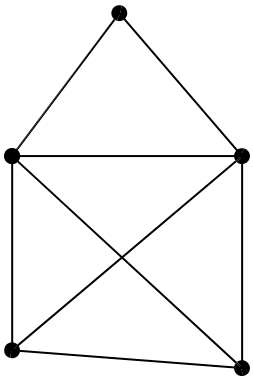


c)



d)

Y



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2) From your answers to question 1, or otherwise, state a rule concerning the number of vertexes of odd degree

3) Draw an example of a planar and a non-planar graph

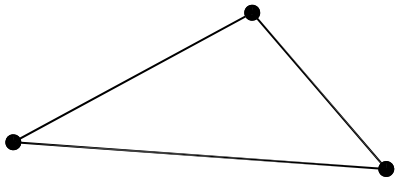
4) For planar graphs, state Euler's formula, and prove the formula holds for your planar graph of question 3

5) The graph in question 1 (c) appears to be non-planar. Redraw the graph to show that any graph with 4 vertices (K_4) is planar

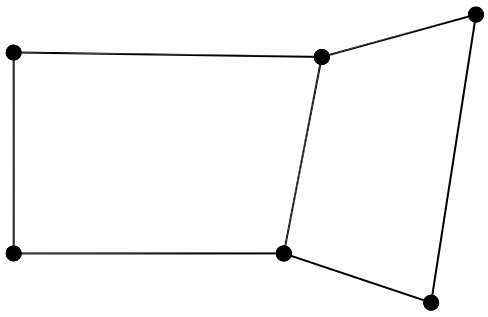
6) Can the graph in question 1 (d) be redrawn to show that it is planar? Explain your answer

7) Which of the following graphs are traversable (that is has at least one Euler path)? Explain your reasons

a)

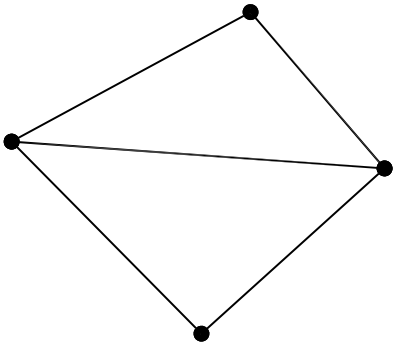


b)

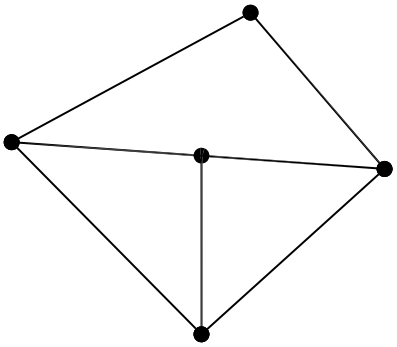


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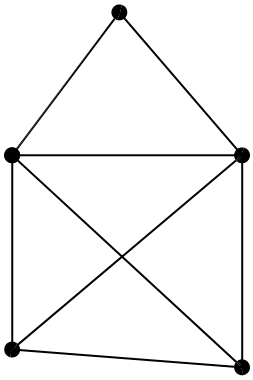
c)



d)

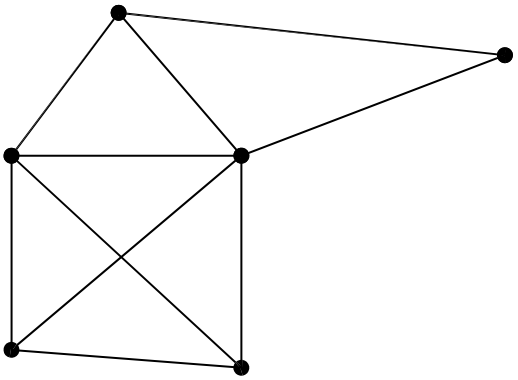


e)

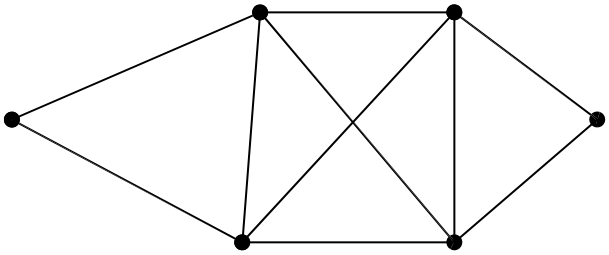


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f)

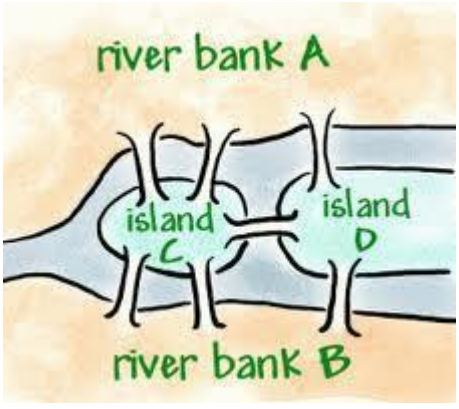


g)



8) Which of the graphs in question 8 is a Hamiltonian path? Explain your answer

9) Below is a simplified representation of the bridges of Konigsberg. By simplifying the diagram further (show as a graph), demonstrate that the network is not traversable



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Exercise 2

Trees

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1) In relation to networks, define the following

a) A tree

b) A spanning tree

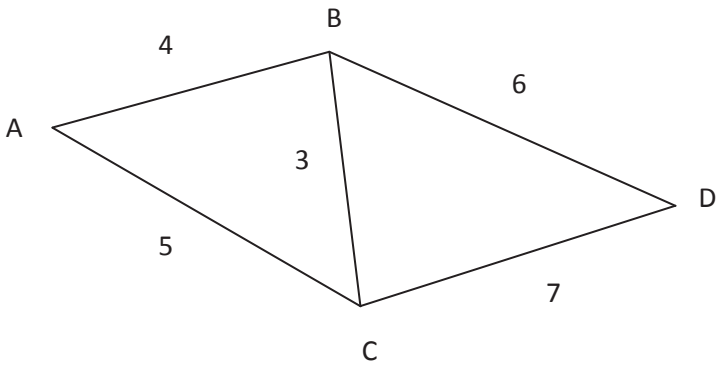
c) A minimum spanning tree

2) What is the objective in solving a minimum spanning tree problem?

3) In a minimum spanning tree problem, what features of a network must be inserted?

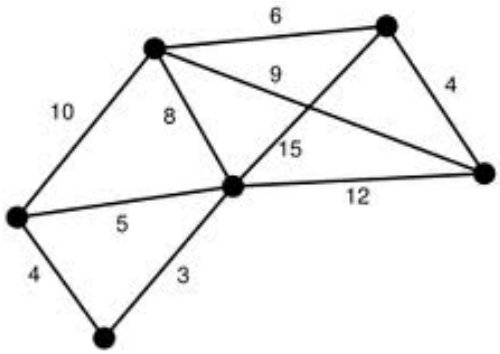
4) List some applications of minimum spanning tree problems

5) The network below shows the cost to connect each node to others. By use of a greedy algorithm, show the minimum spanning tree and hence determine the minimum cost of connecting the network



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1) 6) Assign labels to each node and find the minimum spanning tree for the network



2) The table below shows the distances between various towns

	A	B	C	D	E
A	------	431	531	544	503
B	431	-----	109	120	68
C	531	109	-----	152	105
D	544	120	152	-----	56
E					

a) Complete the table

b) Draw the network for this table

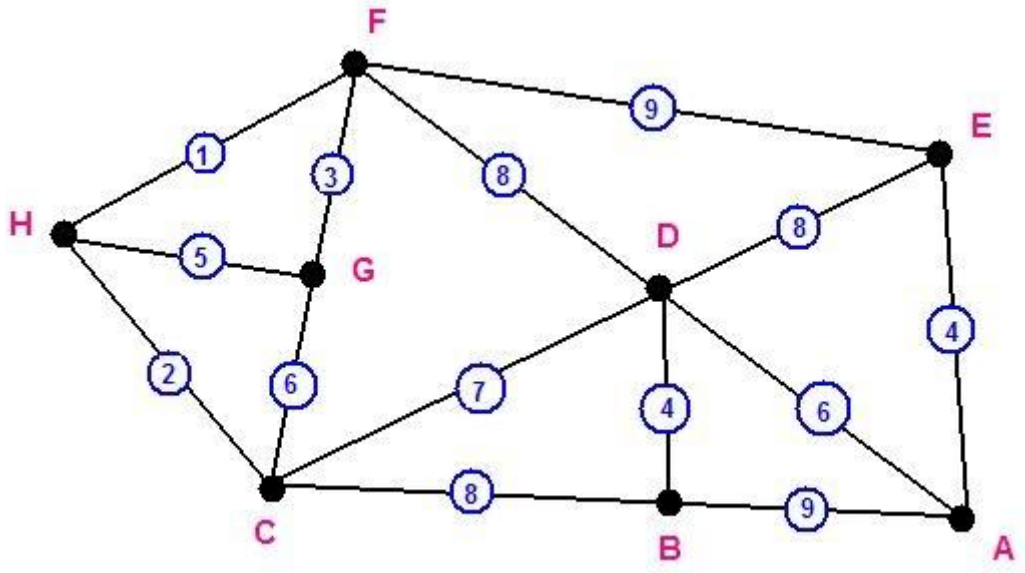
c) Calculate the minimum spanning tree from your diagram

d) Use Prim's Algorithm to calculate the minimum spanning tree

e) Compare the efficiency of using the two methods to find the minimum spanning tree for large networks

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8) Draw a matrix (table) representation of the following network, and use Prim's algorithm to calculate the minimum spanning tree



9) Crazyphones soon will be hooking up computer terminals at each of its branch offices to the computer at its main office, using special phone lines. The phone line from a branch office need not be connected directly to the main office. It can be connected indirectly by being connected to another branch office that is connected (directly or indirectly) to the main office. The only requirement is that every branch office be connected by some route to the main office. The charge for the special phone lines is $50 times the number of kilometres involved, where the distance (in KM) between every pair of offices is as follows:

	Main	A	B	C	D	E
Main	----	190	70	115	270	160
A	190	----	100	110	215	50
B	70	100	----	140	120	220
C	115	110	140	----	175	80
D	270	215	120	175	----	310
E	160	50	220	80	310	----

Management wishes to determine which pairs of offices should be directly connected by special phone lines in order to connect every branch office (directly or indirectly) to the main office at a minimum total cost.

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Use Prim's algorithm to solve the problem. What is the total cost for the special phone lines?

10) A war has devastated most of the internal communications structures of Mathematica, a country on the eastern edge of Asia. The capital of Mathematica is Algebra. The other 6 main cities are Boolean, Complex, Discrete, Euler, Factor, and Graph

The government wishes to re-establish communication between each city and the capital at minimum cost. As long as a city is connected to one other city, and at least one city is connected to the capital all communications will be restored.

The cost to connect cities to each other varies due to distance, conditions and extent of damage caused by the war, and is summarised below

	A	B	C	D	E	F	G
A	----	225,000	165,000	82,000	16,000	310,000	66,000
B		----	222,000	83,000	65,000	145,000	165,000
C			----	100,000	225,000	750,000	315,000
D				----	300,000	185,000	212,000
E					----	120,000	200,000
F						----	150,000
G							----

What is the minimum cost to re-establish connections throughout Mathematica?

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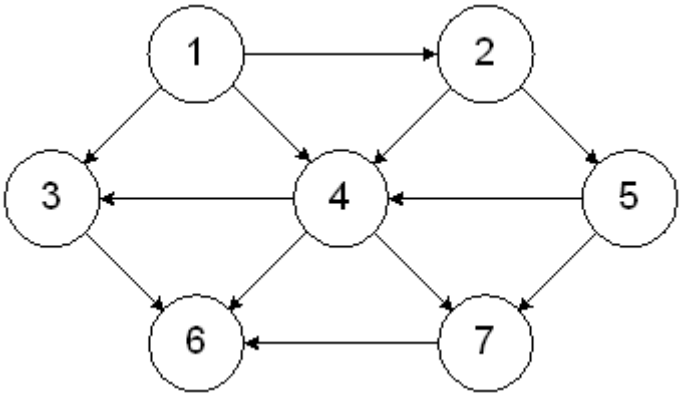
Exercise 3

Paths & Flow

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1) What distinguishes a directed graph from a non-directed graph?

2) What are the indegree and outdegree values of each vertex in the following directed graphs?



3) For the graph in question 2, produce a matrix representation of the one, two and three stage paths between the vertices. Using matrix addition or otherwise, calculate the reachability of each vertex

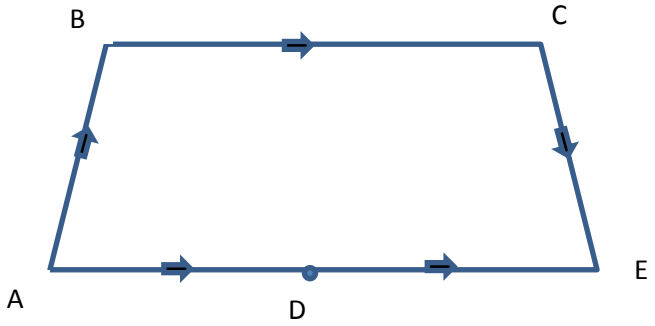
4) A round robin tennis tournament is held for five players who play each other once

- A defeated D, and E
- B lost to A and D
- C did not win a game
- E defeated B and D

a) Construct a graph to represent the outcome of the games

b) Calculate a one-step dominance score and hence rank the players

5) The graph below represents water pipelines between various pumping stations



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The maximum flows in litres per hour between the stations are:

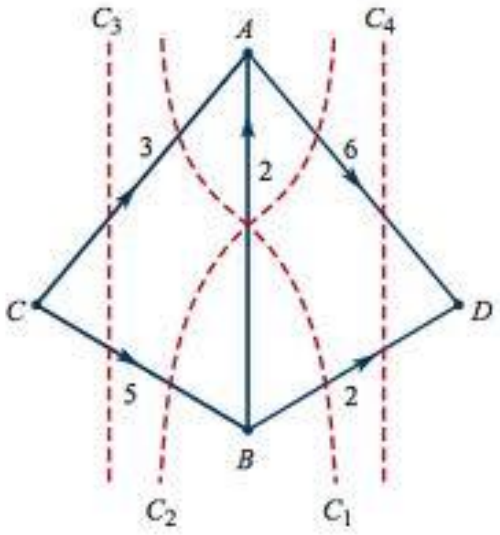
- A to B 300
- B to C 600
- C to E 800
- A to D 500
- D to E 150

a) By inspection, find the maximum flow from A to E through D

b) A new pipe is installed from D to C that has a flow rate of 500. What is the new maximum flow from A to E (not necessarily through D)?

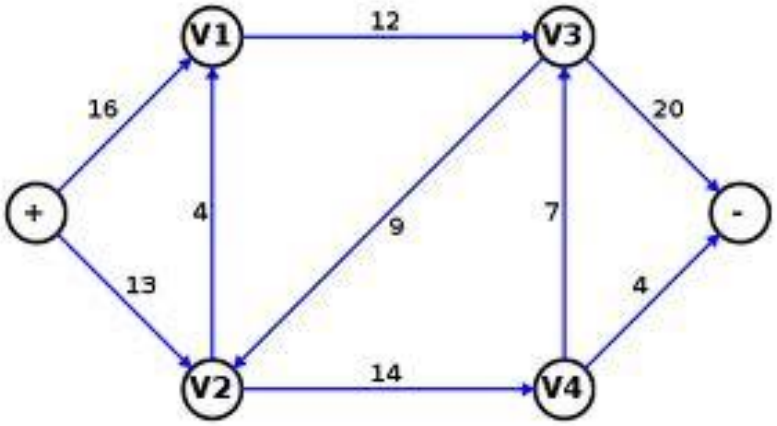
6) Demonstrate use of the minimum cut method to check your answer to question 5 (b)

7) Determine the capacities of each of the cuts in the diagram

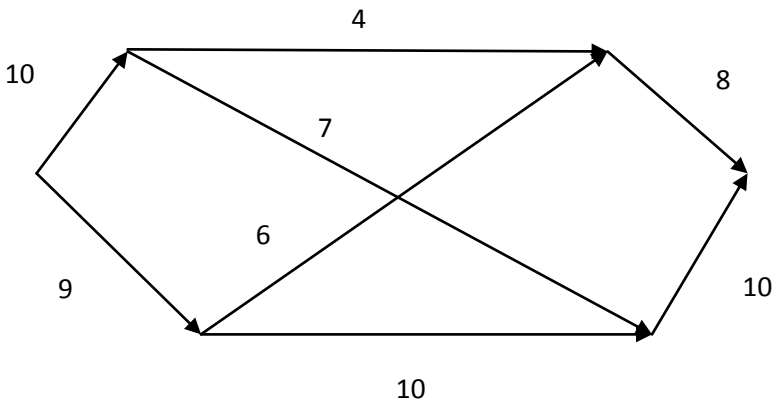


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8) Determine the maximum flow through the following network using the minimum cut method



9) Find the maximum flow for the following network



10) The Department of transport wishes to build a road directly from Alantown to Evanville. Currently there is no direct route between the two towns. A person can still get to Evanville from Alantown, but only via certain routes. The width of the roads, terrain and other factors restrict the maximum number of vehicles that can safely travel between these towns

Alantown links with Badville and Downtown. The first route can carry 800 vehicles per hour, and the second 700

Badville links to Cantown (500 vehicles per hour), and Evanville (400 vehicles per hour)

Cantown also links with Downtown (400 vehicles per hour) and Evanville (800 vehicles per hour)

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Downtown links with Evanville and can transport 200 vehicles per hour
If the government wishes to build a road directly from Alanvtown to Evanville what must be its minimum capacity in order to improve the flow of traffic between the two?

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Exercise 4

Optimisation

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1) Construct a project diagram for the process repair a lawnmower motor

Activity	Description	Immediate predecessors
A	Remove motor	
B	Remove part	A
C	Order part	B
D	Fit new part	C
E	Replace motor	D

2) Complete the following table for replacing a part and repairing a door damaged in an accident, and construct a project diagram

Activity	Description	Immediate predecessors
A	Remove panel	
B	Remove part	A
C	Order part	B
D	Fix dent	A
E	Repaint	D
F	Replace part	C
G	Replace panel	E,F

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3) A project involves nine basic tasks: J, K, L, M, N, O, P, Q, and R. These tasks must be performed by obeying the following sequence rules: Task J must be done before tasks K, L and N. Task L must be done before tasks M and Q. Task O must be done after task K.

Construct a table and project diagram

4) A company produces rubbish bins and needs two machines to make them. Machine A makes the tops of the bins, and both machines are needed to make the bottoms of the bins, which are then assembled and sent out. The activities required, their duration and other details are summarised in the following table

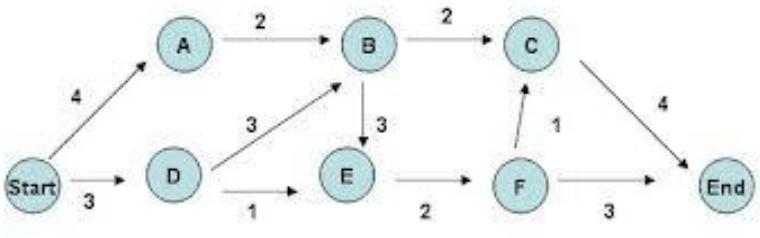
Activity	Description	Estimated duration (days)	Immediate predecessors
A	Purchase and install machine A	8	None
B	Purchase and install machine B	6	None
C	Test machine A	1	A
D	Test machine B	2	B
E	Produce top	3	C
F	Produce bottom	1	D, E
G	Assemble	2	F
H	Ship product	1	G

Construct a diagram and find the critical path and time frame for the production and shipment of a bin

5) Find the earliest and latest starting times for the activities above, and hence identify the critical activities and any float times for non-critical activities

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6) Determine the critical path for the process diagram below, and determine which processes (if any) have float times



7) You are manager of a parts company with offices in Melbourne, Sydney and Adelaide. You wish to send a salesperson from each branch to visit customers in Brisbane, Darwin and Perth. The cost of flying to each city from the three branches is summarised in the table

	Darwin	Brisbane	Perth
Melbourne	700	410	525
Sydney	550	420	480
Adelaide	550	600	375

Where should each salesperson fly in order to minimise airfare?

8) A company has four trucks at 4 different yards and wishes to move each of them to a different sandpit. The distances from their current locations to the sandpits are

	Pit 1	Pit 2	Pit 3	Pit 4
Truck 1	90	75	75	80
Truck 2	35	85	55	65
Truck 3	125	95	90	105
Truck 4	45	110	95	115

To minimise total travel distance, which truck should go to which pit?

9) A company wishes to allocate each of 4 programming jobs to each of 4 programmers. According to experience with each of the programming languages, each programmer will cost the company a different amount to do each job. This is summarised below with the cost in hundreds of dollars

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	Java	HTML	C++	PHP
Alan	80	105	95	100
Jill	25	85	65	95
Peter	105	75	60	110
Boris	145	100	90	120

To minimise total cost, which programmer should get which job?

10) A company has 5 cleaning tasks and 5 cleaners to do the job. The company has assigned an "efficiency rating" for each cleaner for each task, based on previous performance etc. The higher the rating the more efficient the cleaner is at that particular task. Which cleaner should do which job in order to MAXIMIZE efficiency? (no job is dependent on any other; they can all be started at the same time)

	Mop	Windows	Rubbish	Desks	Amenities
Bob	10	19	8	15	16
Carol	10	18	7	17	12
Glen	13	16	9	14	8
Jason	12	19	8	18	11
Rachel	14	17	10	19	15

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Further Mathematics

Matrices

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Exercise 1

Representation & Operations

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1) What will be the dimensions of the matrix formed by multiplying matrices of the following dimensions?

a) $(2 \times 3) \times (3 \times 2)$

b) $(1 \times 4) \times (4 \times 1)$

c) $(3 \times 3) \times (3 \times 5)$

d) $(1 \times 3) \times (3 \times 2)$

e) $(2 \times 3) \times (1 \times 2)$

2) Perform the following operations giving your results in matrix form

a) $\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} + \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$

b) $\begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} -3 & 0 \\ -1 & 1 \end{bmatrix}$

c) $2\begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ -1 & -2 \end{bmatrix}$

d) $\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -4 & 4 \end{bmatrix} - 2\begin{bmatrix} 1 & -\frac{1}{2} \\ -1 & 1 \end{bmatrix}$

e) $\begin{bmatrix} 84 & 763 \\ 1234 & -499 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

3) Multiply the following matrices

a) $\begin{bmatrix} 1 & 2 \\ -2 & -1 \end{bmatrix} \times \begin{bmatrix} -2 & 1 \\ 2 & -1 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \times \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} -2 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix}$

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d) $\begin{bmatrix}1 & 2\\4 & 3\end{bmatrix}\times\begin{bmatrix}1 & 1 & 1\\1 & 1 & 1\\1 & 1 & 1\end{bmatrix}$

4) Given the following matrices, perform the operations if possible. If the operation is not possible state the reason

$$A=\begin{bmatrix}2 & -3 & 4\\3 & 0 & 1\\2 & -2 & 1\end{bmatrix}\quad B=\begin{bmatrix}3 & -2 & 1\\1 & 4 & -1\\-1 & 2 & 1\end{bmatrix}\quad C=\begin{bmatrix}2 & 4 & 8\end{bmatrix}\quad D=\begin{bmatrix}-1\\4\\2\end{bmatrix}$$

a) 4C

b) AD

c) DA

d) BC

e) 3CB

f) C(A+B)

g) AB

h) BA

i) CAD

j) DBC

5) The organisers of a sports competition need to supply lunches to the teams. The choices are a roll, fruit and a drink for each competitor. A competitor can have one, two or all of the choices. Each team submitted their requirements to the organisers. On day 1, team red required 10 burgers, 8 pieces of fruit and 7 drinks; team blue wanted 12, 6, and 9; and team green ordered 11, 10 and 5.

Represent the requirements for each team for day 1 in a matrix

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6) On the second day the orders were (rolls, fruit, drink); red 12, 6, 8; blue 9, 9, 7; green 11, 11, 8

Represent the requirements for day 2 in a separate matrix

7) On the third day each team also had a reserves side play. To make life easier for the organisers, the teams simply doubled their order from day 2.

Represent the requirements for day 3 in a separate matrix and show the operation used to generate it

8) Show the operation required to calculate the numbers of rolls fruit and drinks for each team for the three days of the tournament, and hence calculate the totals and show in matrix form.

9) Each roll cost $3.50, each piece of fruit cost 75 cents, and each drink cost $2.45.

a) Construct a one column matrix to represent these prices

b) How much did the red team spend on fruit?

c) How much did the green team spend on drinks?

d) Use matrix multiplication to show the total costs for each team for the three days

10) The matrix A represents the average score for 5 students in tests quizzes and homework. Tests contribute 60% toward a student's final grade, quizzes are worth 15% and homework 25%

	T	Q	H
Peter	78	80	75
Brett	76	90	95
Amy	72	70	85
Karen	60	70	80
Sue	84	80	90

$= A$

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a) Write a vector R that would represent the weighting given to each component of a student's score

b) Would the matrix AR or RA be used to calculate the final weighted grade for each student? Explain

c) Calculate the final weighted grade for each student

11) The hire charges for a bus from three different companies is based on the number of days for which a bus is hired and/or the number of kilometres for which the bus is driven. The rates from the three companies are:

Company A charges $66 per day.
Company B charges 48 cents on per kilometre driven.
Company C charges $30 per day and 25 cents on per kilometre driven.

Alan needs to hire a bus for 4 days to drive 560 kilometres

Write two matrices that represent the above information, and use matrix multiplication to find which company Alan should hire the bus from

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Exercise 2

Simultaneous Equations

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1) Calculate the inverse of the following matrices

a) $\begin{bmatrix} 4 & -1 \\ 4 & 1 \end{bmatrix}$

b) $\begin{bmatrix} 2 & 6 \\ 0.5 & 3 \end{bmatrix}$

c) $\begin{bmatrix} 1 & 3 \\ 5 & 6 \end{bmatrix}$

d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

e) $\begin{bmatrix} 6 & 4 \\ 3 & 2 \end{bmatrix}$

2) Represent the following equations in matrix form

a) $2x + y = 4, x - y = 2$

b) $3x + 2y = 7, x + 3y = 1$

c) $2x = y - 3, y + 2x = -1$

d) $x = y, 2x = y - 4$

e) $x = 3, y = 4$

f) $x = y, x = -y$

3) Solve the following simultaneous equations by use of matrices

a) $x + y = 3, 2x - y = 0$

b) $2x + y = 5, x - y = -2$

c) $x = 4 - 2y, y = -x + 2$

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d) $2y - 3x = -8,\ 2x = 4y + 4$

e) $x + \frac{1}{2}y = 4, 4x + 2y = 3$

4) The sum of two numbers is 6, and their difference is 4. What are the numbers?

5) The length of a rectangle is 3 times its width. The perimeter of the rectangle is 40 cm. Find its dimensions.

6) The difference between two numbers is 2. The sum of three times the larger number and twice the smaller is 11. What are the numbers?

7) 5 books and 2 pens cost $9, while 2 books and 10 pens cost $8.20. What is the cost of a book and a pen?

8) The sum of the numerator and denominator of a fraction is 7. If the denominator is increased by 3, the fraction becomes $\frac{1}{4}$. What is the original fraction?

9) Tom is Kevin's father; Twice Kevin's age plus Tom's age equals 58. In 5 years the sum of their ages will be 56. How old are Tom and Kevin currently?

10) Two runners start from the same point at the same time. If they run in the same direction they will be 2 km apart after 2 hours. If they run in opposite directions they will be 14 km apart after 2 hours. At what speed does each runner travel?

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Exercise 3

Transition Matrices

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1) For matrix A and B given below, calculate

a) AB

b) AB^2

c) AB^3

d) AB^5

e) AB^0

$$A = \begin{bmatrix} 0.8 & 0.6 \\ 0.2 & 0.4 \end{bmatrix} \quad B = \begin{bmatrix} 0.5 & 0.4 \\ 0.5 & 0.6 \end{bmatrix}$$

2) John and Ken are playing tennis. In their first game each has an equal chance of winning. If John wins he gets more confident and his chance of winning the next game increases to 60%, but if he loses his chance of winning the next game drops to 30%

a) Determine the probability that John wins the second game

b) What are each player's probabilities after 5 games?

3) At the end of 2011 there were 3.1 million people living in the metropolitan area of a state, and 800,000 living in the country. Each year 4% of the people from the city move to the country, and 8% of people in the country move to the city.

a) Write the transition matrix for the movement of people in any one year

b) What are the respective populations at the end of 2012?

c) If the trend continues, what will be the respective populations at the end of 2015?

4) Cools supermarkets currently has 40% of market share, and Woolless supermarkets have 20% of the market share. Each year 10% of Cools customers switch to Woolless, and 5% of Woolless customers switch to Cooles. If the current market is 100,000 families, what will be their respective market shares in 5 years?

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5) For the scenario in question 4, when will the two companies have the same market share and what will this be?

6) At the start of the day there are 60 taxis at the airport and 20 in the city. Each hour 75% of the taxis at the airport travel to the city, and 40% of the taxis in the city travel to the airport. What will be the distribution of taxis after 4 hours?

7) There are three locations for courier services: City, Airport and Suburbs. From the city, couriers stay in the city 50% of the time and go to the airport and suburbs 25% and 25%. From the airport couriers stay at the airport 30% of the time and go to the city and to the suburbs at the rates of 50% and 20% respectively. From the suburbs couriers stay in the suburbs 10% of the time and go to the city and to the airport at the rates of 55% and 35% respectively.

Find the distribution of couriers after one job if they start out with 30 couriers in the city, 20 at the airport; and 10 in the suburbs

8) The following transition matrix shows the movement of people between 4 towns; Calisto, Europa, Ganymede, and Titan. At the beginning of 2010 their populations were 10,000 20,000 30,000 and 40,000 respectively

$$\begin{matrix} C & E & G & T \end{matrix}$$
$$\begin{bmatrix} 0.95 & 0 & 0.01 & 0.04 \\ 0.01 & 0.98 & 0.01 & 0 \\ 0.02 & 0.01 & 0.97 & 0 \\ 0.02 & 0.02 & 0 & 0.96 \end{bmatrix} \begin{matrix} C \\ E \\ G \\ T \end{matrix}$$

a) What will be the steady state population of Titan?

b) Eventually what will be the population of each town?

c) What will the populations be in 2013?

d) What were the populations in 2009?

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